

No Primitive Lower Higgs Vacuum

Endpoint-Standing Collapse, Electroweak Crossover, and Minimal-SM EWBG Route Deletion under AASC

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June 2026

Abstract

The lower Higgs vacuum is not a primitive same-scope endpoint of the measured minimal Standard Model. A lower branch of a gauge-, scheme-, scale-, and matching-dependent effective-potential representative is an admissible diagnostic, but endpoint standing begins only after the branch descends to a quotient-stable stability, lifetime, or richer-scope source certificate. The result does not deny the diagnostic use of RG-improved effective potentials or certificate-dependent metastability claims. It denies the promotion of a lower effective-potential branch, as such, into primitive same-scope endpoint standing. High-scale minimal-SM continuation is treated as admissibility-relevant invariant-bundle content for lifetime verdicts, not as new physics and not as primitive branch standing. The paper further proves that same-domain route availability for C_{meta} is exhausted in form. Every surviving route-available metastability protocol in the fixed target factors through the lifetime-route normal form $\mathcal{R}_{\text{life}}$; route availability itself neither generates endpoint standing nor revives primitive lower-branch standing. Standard metastability calculations are therefore classified rather than rejected: candidate protocols are assessed by route status, route-normal protocols are route-available, and only a closed singleton/lifetime certificate makes C_{meta} standing-bearing.

AASC is used as a mathematical constraint formalism for endpoint standing. Standard quantum field theory supplies the comparison arena: measured Higgs/top anchors, Nielsen-identity gauge control, RG and matching transport, high-scale vacuum-stability analyses, false-vacuum decay, finite-temperature electroweak thermodynamics, and electroweak-baryogenesis route conditions. The AASC contribution is the status calculus that separates diagnostic availability, quotient endpoint standing, singleton-certificate selection, same-scope deletion, route-form exhaustion, and richer-scope update.

The theorem set has three endpoint outputs, together with one route-form exhaustion result:

$$\text{Surv}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0, \quad \text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1, \quad \text{Surv}_{D_{\text{EWBG},0}}(\text{EWBG}_{\text{SM}}) = 0.$$

The accompanying route-form result separates candidate route protocols from route availability. For $D = D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]$,

$$\text{CandRoute}_D(C_{\text{meta}}; \mathcal{P}) \Rightarrow \rho_{\text{route}}^D(\mathcal{P}) \in \left\{ \begin{array}{l} \text{DiagnosticOnly, RouteIncomplete,} \\ \text{RouteNormal, TargetUpdate,} \\ \text{StandingSmuggling} \end{array} \right\},$$

and

$$\begin{aligned} \text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1 &\Rightarrow \text{NF}_{\text{life}}(\mathcal{P}) \downarrow \\ &\wedge \mathcal{P} \cong_D \text{NF}_{\text{life}}(\mathcal{P}). \end{aligned}$$

This is a statement about same-domain route grammar, not standing closure. Thus the fixed measured Higgs target deletes primitive lower-branch standing while preserving metastability as a

certificate-dependent endpoint class. The measured minimal Standard Model finite-temperature target selects crossover and deletes the same-scope first-order branch. The same-scope minimal-Standard-Model electroweak-baryogenesis route, understood as a successful observed-asymmetry production-and-preservation route, has no certified survivor because the required electroweak departure and adequate CP/source limbs are absent. The result does not deny conditional metastability calculations; it denies only the primitive claim-form that a raw lower branch itself has endpoint standing. Any successful lower-vacuum endpoint, first-order transition, or electroweak-baryogenesis route requires a richer-scope or singleton certificate with its own carrier, transport, descent, and witness burdens.

Keywords: Higgs false vacuum; primitive endpoint deletion; vacuum stability; metastability; route availability; route normal form; route exhaustion; high-scale continuation; invariant bundle; effective potential; Nielsen identity; renormalization group; electroweak crossover; electroweak baryogenesis; Standard Model; AASC; zero-point energy.

Mathematical-status lock and proof-class boundary

AASC is used here as a mathematical constraint formalism. It is not a Higgs-sector model, a replacement for RG-improved vacuum-stability calculations, a new lattice computation, a naturalness prior, or an interpretive overlay on an otherwise completed endpoint act. Standard quantum field theory supplies the comparison arena: the Higgs effective potential, Nielsen-identity gauge control, RG/scheme and matching transport, the measured Higgs/top/strong-coupling anchor ledger, false-vacuum decay, finite-temperature electroweak thermodynamics, and the minimal-Standard-Model electroweak-baryogenesis route facts. AASC supplies the status calculus by which formal availability, diagnostic use, endpoint standing, declared input, empirical witness, richer-scope update, and primitive promotion are separated.

The proof is the theorem chain stated in this manuscript. A same-target objection must identify a failure in a definition, verdict map, quotient kernel, target split, lower-branch no-fifth-function theorem, route-availability exhaustion theorem, singleton-certificate burden, imported crossover theorem, imported measured minimal-SM electroweak-baryogenesis failure theorem, route-deletion corollary, or final survivor inference. A numerical instability-scale computation is theorem-relevant only if it supplies, breaks, or updates one of the declared verdict certificates. A demand for a different proof class, a model-building preference, or a phenomenological recalculation does not contest the same endpoint theorem unless it breaks a load-bearing link in the present chain.

Read this first: endpoint proof spine

The proof has five layers.

Layer I: kernel-forced endpoint use. A determinate Higgs-vacuum endpoint claim already requires reference, standing, admissibility, and irreversibility:

$$\text{EndpointUse}_{D_{\text{HV},0}}(C) \Rightarrow K_D = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}).$$

A raw effective-potential representative, field-coordinate lower branch, running-coupling crossing, thermal profile, or CKM-only source cannot define the standing rule by which it wins.

Layer II: target, candidates, and verdict maps. The fixed targets are

$$D_{\text{HV},0}^{T=0}[\text{S}_{2026}], \quad D_{\text{HV},0}^{T>0}, \quad D_{\text{EWBG},0},$$

with richer-scope updates $D_{\text{HV},+}$ and $D_{\text{EWBG},+}$. The lower branch is admitted as diagnostic content. Endpoint-bearing use must factor through the verdict maps

$$\nu_{\text{stat}}, \quad \nu_{\text{life}}, \quad \nu_{\text{therm}}$$

or through target update. Quotient objects are kernels of those verdict maps, not raw field-space pictures. For lifetime verdicts, the high-scale minimal-SM continuation clause belongs to the invariant bundle I_D : it fixes the same target and same admissibility boundary for RG transport, lower-branch diagnostics, and bounce/lifetime evaluation.

Layer III: no primitive lower-vacuum endpoint. The proof does not use the shortcut

$$\text{LowerBranch}(V_{\text{eff}}) \Rightarrow \text{PrimitiveStanding}(C_{\text{lower}}).$$

The typed route is

$$V_{\text{eff}}^{(g,s,\mu,\mathcal{M})} \rightarrow B \rightarrow \text{DiagnosticAvailability} \rightarrow \{\nu_{\text{stat}}, \nu_{\text{life}}, \nu_{\text{therm}}, D_{\text{HV},+}\} \rightarrow \text{route exhaustion}.$$

The no-fifth-function theorem closes the route family. More explicitly,

$$\neg \exists F_5(B) \wedge \text{Exhaust}(\nu_{\text{stat}}, \nu_{\text{life}}, \nu_{\text{therm}}, D_{\text{HV},+}) \Rightarrow \text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0.$$

Thus

$$\text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0.$$

Layer IV: route-availability exhaustion. The paper also exhausts same-domain route availability for metastability. A candidate protocol $\mathcal{P}$ that purports to make C_{meta} available inside $D = D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$ is first classified by a fail-closed route-status classifier:

$$\text{CandRoute}_D(C_{\text{meta}}; \mathcal{P}) \Rightarrow \rho_{\text{route}}^D(\mathcal{P}) \in \left\{ \begin{array}{l} \text{DiagnosticOnly, RouteIncomplete,} \\ \text{RouteNormal, TargetUpdate,} \\ \text{StandingSmuggling} \end{array} \right\}.$$

Route availability is reserved for the surviving route-normal case:

$$\begin{aligned} \text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1 &\Rightarrow \text{NF}_{\text{life}}(\mathcal{P}) \downarrow \\ &\wedge \mathcal{P} \cong_D \text{NF}_{\text{life}}(\mathcal{P}). \end{aligned}$$

Thus same-domain route availability has a fixed lifetime normal form, while route availability remains distinct from standing closure. No further same-domain route-availability class remains.

Layer V: positive thermal endpoint and same-scope measured minimal-SM route deletion. The finite-temperature measured minimal Standard Model selects crossover:

$$\text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1, \quad \text{Surv}_{D_{\text{HV},0}^{T>0}}(C_{\text{1st}}) = 0.$$

The same-scope minimal-Standard-Model electroweak-baryogenesis route has no certified survivor:

$$\text{Surv}_{D_{\text{EWBG},0}}(\text{EWBG}_{\text{SM}}) = 0.$$

A lower-vacuum endpoint, first-order transition, or successful electroweak-baryogenesis route can re-enter only by a specified singleton or richer-scope certificate.

Dependency-inversion exclusion

The dependency order is not

AASC framework $\Rightarrow$ Higgs endpoint conclusion.

The dependency order is

determinate fixed Higgs endpoint evaluation $\Rightarrow K_D \Rightarrow$ AASC consequence layer.

A candidate branch cannot define the selector by which it is admitted as endpoint standing. Once the target is a determinate comparison among lower-branch, stability, lifetime, thermal, and electroweak-baryogenesis claims, target identity, branch evaluability, lawful redescription, and act-time finality are already required. A same-target construction that denies those roles must either derive them from a lower non-circular basis, concede governance-equivalent work under another vocabulary, or abandon the fixed Higgs-vacuum target.

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Executive result and claim classes

The endpoint theorem has three selected endpoint outputs:

$$\text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0, \quad \text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1, \quad \text{Surv}_{D_{\text{EWBG},0}}(\text{EWBG}_{\text{SM}}) = 0.$$

The route-availability extension also proves one route-form exhaustion result inside the fixed zero-temperature target. Candidate protocols are first classified, and route availability is reserved for route-normal survivors:

$$\begin{aligned} \text{CandRoute}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) &\Rightarrow \\ \rho_{\text{route}}^{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(\mathcal{P}) &\in \left\{ \begin{array}{l} \text{DiagnosticOnly, RouteIncomplete, RouteNormal,} \\ \text{TargetUpdate, StandingSmuggling} \end{array} \right\}. \\ \text{RouteAvail}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) = 1 &\Rightarrow \text{NF}_{\text{life}}(\mathcal{P}) \downarrow \\ &\wedge \mathcal{P} \cong_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]} \text{NF}_{\text{life}}(\mathcal{P}). \end{aligned}$$

This records route-form normality rather than standing closure. The proof spine has already fixed the dependency order. This section records the claim classes used below: standard Higgs/electroweak field theory supplies the comparison arena; AASC supplies the status calculus; theorems select or delete only certified same-target survivors.

Class	Role in this manuscript
Standard field-theory input	Effective potentials, Nielsen-identity gauge control, RG/matching transport, declared high-scale minimal-SM continuation, false-vacuum decay structure, physical-Higgs crossover, and measured minimal-SM EWBG failure facts.
AASC status theorem	Target fixation, availability/standing distinction, no-promotion, verdict-map quotienting, route exhaustion, and same-target counterexample discipline.
Endpoint theorem	Selection or deletion over certified same-target survivors: primitive lower-branch deletion, crossover selection, first-order deletion, and same-scope measured minimal-SM EWBG route deletion.
Scope/update theorem	Recognition of singleton certificates, source-state updates, richer-scope Higgs updates, and richer-scope EWBG updates as target changes rather than same-scope counterexamples.

1 Introduction: the lower Higgs vacuum as a status problem

The lower Higgs vacuum is usually introduced through a picture: the Standard Model effective potential, RG-improved to very high field values, appears to develop a lower region for central values of the Higgs, top, and strong-coupling inputs. In ordinary language this is often compressed into the statement that the electroweak vacuum may be false. The AASC result is more precise: the lower branch is not an endpoint. It is a diagnostic feature of a transported representative. A physical false-vacuum endpoint exists only after a quotient-stable lifetime or stability certificate is supplied.

Throughout, the source-state symbol S_{2026} denotes the evidence and anchor ledger used in this manuscript—the measured inputs, published high-order analyses, and declared AASC source dependencies available to the target. It is mutable by design: a later precision or carrier certificate creates a new source state rather than refuting the theorem proved for S_{2026} . A *primitive endpoint* means a claim used as endpoint standing without first being derived from a quotient-stable stability, lifetime, or richer-scope certificate.

The paper therefore distinguishes three objects that are often collapsed:

$$C_{\text{lower}} \neq C_{\text{meta}} \neq \text{LowerBranch}(V_{\text{eff}}^{(g,s,\mu,\mathcal{M})}).$$

C_{lower} denotes the candidate primitive lower-vacuum branch. C_{meta} denotes a metastability endpoint with a lifetime verdict. $\text{LowerBranch}(V_{\text{eff}})$ denotes a representative diagnostic. The main zero-temperature endpoint theorem deletes the first object as primitive standing while preserving the second as a certificate-dependent endpoint class whose standing requires singleton/lifetime closure.

The phrase “without new physics” is therefore not rejected; it is typed. A high-scale continuation assumption stating that the minimal Standard Model remains the declared carrier inventory up to the declared protocol support envelope adds no new particles or forces. It is nevertheless boundary content for the endpoint act: it fixes the domain over which RG transport, matching, and lifetime evaluation are claimed to remain same-scope. Thus no-new-physics continuation can support a certified ν_{life} route to C_{meta} , but it cannot by itself promote $\text{LowerBranch}(V_{\text{eff}})$ to C_{lower} with primitive standing.

The paper also exhausts the route-availability side of this distinction. A same-domain minimal-SM metastability route is not any theoretical pipeline that happens to mention a lower branch; it must factor through

$$\mathcal{R}_{\text{life}} = (A_{\text{HV}}, I_D, \text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}}), T_{\text{RG}}, B, \mathcal{L}, U_{\Delta}, S).$$

Thus existing metastability calculations are admitted as route candidates when their measured anchors, invariant-bundle content, declared high-scale continuation, RG/matching transport, gauge-controlled branch diagnostic, lifetime package, uncertainty/separation envelope, and source state are declared. Their availability is not standing closure and does not revive primitive standing for C_{lower} .

This distinction is forced by the structure of gauge theory and RG transport. In a gauge theory the effective potential is not a raw gauge-invariant scalar function of a field coordinate. The Nielsen identity gives the schematic relation

$$\left(\xi \frac{\partial}{\partial \xi} + C(\phi, \xi) \frac{\partial}{\partial \phi} \right) V(\phi, \xi) = 0,$$

so gauge variation of a potential representative is tied to field-coordinate displacement. Values at properly treated extrema can be gauge controlled, but field coordinates, instability field values, and representative lower-branch locations cannot be read directly as physical endpoints. Di Luzio and Mihaila distinguish gauge-independent stability bounds from gauge-dependent instability field values [9], and Andreassen, Frost, and Schwartz emphasize that physical use of effective potentials requires consistent perturbative and gauge-aware extraction [10].

The finite-temperature side has a firmer same-scope endpoint. With the physical Higgs mass, lattice and thermodynamic analyses classify the Standard Model electroweak transition as a smooth crossover rather than a first-order transition [13, 14]. This deletes the same-scope first-order electroweak phase-transition branch. It also deletes the same-scope minimal-Standard-Model electroweak-baryogenesis route: the first-order bubble-wall route is absent, and CKM-only CP violation is not an adequate electroweak-baryogenesis source [19, 20].

At zero temperature the result is different. Absolute stability, metastability, and instability are legitimate endpoint classes. The source-state anchor ledger, however, does not structurally select a singleton. Source-state analyses still identify the top mass and strong coupling as load-bearing for Standard Model vacuum stability, with improved precision needed to establish or refute stability at high significance [6]. The top-mass input is itself a transport issue because direct measurements quote a Monte-Carlo mass parameter whose field-theoretic interpretation is nontrivial [3].

The theorem set is therefore:

1. primitive lower-Higgs-vacuum standing is deleted;
2. candidate metastability routes are exhausted by the ordered route-status classifier, and route availability survives only through NF_{life} as route form, not standing closure;
3. finite-temperature electroweak crossover is selected;
4. same-scope first-order transition is deleted;
5. same-scope minimal-SM electroweak baryogenesis is deleted;
6. zero-temperature stability/metastability selection requires an anchor-closed singleton certificate that the source state does not itself provide.

2 Constraint kernel, target lock, and status calculus

2.1 AASC as a constraint formalism

AASC is used here as a fixed-target status calculus. It does not add particles, alter the Standard Model, choose a top-mass scheme, compute a bounce action, run a lattice simulation, or fit a new Higgs self-coupling. Its proof tools are target fixation, quotient construction, no-promotion, selector-burden classification, route exhaustion, and richer-scope recognition.

The external physics inputs are standard: Higgs and top mass measurements, RG-improved Standard Model vacuum-stability analyses, Nielsen-identity gauge control of effective potentials, finite-temperature electroweak thermodynamics, and known BSM routes to first-order transitions. AASC audits the inferential status of claims made from those inputs. The theorem is therefore not a request that high-energy theory adopt AASC vocabulary before the physics can be discussed. It is a fixed-target consequence theorem: once Higgs-vacuum endpoint claims are made, the target, candidate classes, admissible transports, endpoint verdicts, and update conditions must be governed somehow. AASC names and audits that governance work.

2.2 Theorem-class lock

Proposition 2.1 (Mathematical-status and same-target admissibility lock). *The theorem is evaluated inside a declared proof calculus: AASC plus the standard Higgs/electroweak/QFT comparison arena. A same-target objection must invalidate a definition, a quotient construction, a transport rule, a dependency claim, an endpoint-selection criterion, a physics-to-status bridge, or the final survivor inference; alternatively it must derive the relevant kernel role from a strictly lower non-circular basis or exhibit a faithful same-regime counterexample.*

Proof. Every theorem is conditional on a proof setting. Dependence is a defect only if the dependency is false, misapplied, circularly smuggled into the target, or insufficient for the conclusion. A change of vocabulary or a preference for a naturalness formulation is not by itself a same-target objection. A numerical instability-scale calculation is same-target relevant only when it supplies, breaks, or

updates a declared verdict certificate. To contest the present theorem, the objection must preserve the Higgs-vacuum target and attack a load-bearing link in the endpoint chain. $\square$

2.3 Denial obligations

A denial of the endpoint theorem has standing only if it does at least one of the following:

1. shows that claim-bearing Higgs-vacuum endpoint use does not require target identity, status evaluability, same-regime fidelity, or act-time finality;
2. derives the corresponding governance role from below without presupposing equivalent governance work;
3. identifies a failure in the typed quotient construction $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, or $[\mathcal{F}_T]_{T>0}^{\text{therm}}$;
4. supplies a same-target lower-branch endpoint with primitive standing independent of quotient audit;
5. breaks the candidate-route classifier, route-status completeness theorem, or lifetime-route normal-form exhaustion theorem;
6. supplies a closed zero-temperature singleton certificate $\mathcal{S}_{\text{HV}}^{T=0}(C)$ from the source-state anchors alone;
7. produces a same-scope first-order electroweak transition in the measured minimal Standard Model; or
8. supplies a faithful same-regime counterexample satisfying the preservation conditions in Section 19.2.

Absent one of these results, the objection either changes target, imports hidden selector work, treats a representative as the endpoint, or refuses terminology while retaining the same governance work.

2.4 What the paper proves

This paper proves a theorem set over the fixed measured minimal Standard Model Higgs-vacuum target $D_{\text{HV},0}$:

$$\begin{aligned}
& \text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1, \quad \text{Surv}_{D_{\text{HV},0}^{T>0}}(C_{1\text{st}}) = 0, \\
& \text{Surv}_{D_{\text{EWBG},0}}(\text{EWBG}_{\text{SM}}) = 0, \\
& \text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0, \\
& \text{LowerBranch}(V_{\text{eff}}^{(g,s,\mu,\mathcal{M})}) \not\Rightarrow \text{PrimitiveStanding}_{D_{\text{HV},0}^{T=0}}(C), \\
& \text{Sel}_{D_{\text{HV},0}^{T=0}}(C_{\text{abs}}) \vee \text{Sel}_{D_{\text{HV},0}^{T=0}}(C_{\text{meta}}) \Rightarrow \exists \mathcal{S}_{\text{HV}}^{T=0}(C), \\
& \neg \exists \mathcal{S}_{\text{HV}}^{T=0}[\text{S}_{2026}](C_{\text{abs}}) \wedge \neg \exists \mathcal{S}_{\text{HV}}^{T=0}[\text{S}_{2026}](C_{\text{meta}}).
\end{aligned}$$

The source-state nonexistence claim is not a statement that no numerical analysis favors metastability. It is the structural endpoint claim that the measured-anchor ledger, by itself, does not close all selector-burden components required for an AASC singleton.

2.5 What the paper does not prove

Scope statement 2.2 (No new numerical instability scale). A numerical instability-scale computation belongs to the present theorem only when it supplies, breaks, or updates a declared verdict certificate: ν_{stat} , ν_{life} , a singleton certificate $\mathcal{S}_{\text{HV}}^{T=0}(C)$, or a source-state update. A calculation that merely displays a representative scale remains diagnostic availability rather than primitive endpoint standing.

Scope statement 2.3 (No structural metastability singleton). The paper does not select metastability as a purely structural singleton. It preserves metastability as a legitimate endpoint class requiring lifetime and anchor-closed certification.

Scope statement 2.4 (No anti-BSM result). The paper does not rule out singlet scalars, two-Higgs-doublet models, supersymmetry, scalar portals, SMEFT barriers, or gravitational corrections. It classifies them as $D_{\text{HV},+}$ target updates when their carrier and descent certificates are specified.

3 Fixed measured Higgs-vacuum target

Definition 3.1 (Fixed measured minimal Standard Model Higgs-vacuum target). $D_{\text{HV},0}$ is the fixed measured minimal Standard Model Higgs-vacuum comparison target containing:

1. the Standard Model gauge and matter carrier inventory;
2. a single Higgs doublet with electroweak bridge role;
3. the measured Higgs-sector anchor;
4. the measured top-sector anchor and its mass-scheme interpretation burden;
5. the strong-coupling input required for RG evolution;
6. admissible perturbative/RG-improved matching machinery;
7. the zero-temperature effective-potential and decay/lifetime comparison arena;
8. the finite-temperature electroweak thermodynamic comparison arena;
9. no additional scalar, portal, threshold, supersymmetric, gravitational, asymptotic-safety, or UV-stabilizing carrier unless declared as richer scope.

The target contains the arena and candidate classes, not a preselected endpoint.

The modern anchor values make the problem sharply measured. ATLAS reports a combined Run 1+Run 2 Higgs mass of $m_H = 125.11 \pm 0.11$ GeV [1]. The ATLAS+CMS Run-1 top-mass combination reports $m_t = 172.52 \pm 0.33$ GeV [2]. These values anchor the target. They do not themselves collapse the endpoint-status problem.

Proposition 3.2 (Context non-smuggling). *The fixed measured Higgs targets $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$, $D_{\text{HV},0}^{T>0}$, and $D_{\text{EWBG},0}$ fix carrier inventory, measured anchors, admitted representatives, verdict maps, route candidates, and update protocols only. They do not assert*

$$\text{Surv}^{\text{prim}}(C_{\text{lower}}) = 0, \quad \text{Sel}(C_{\text{cross}}) = 1, \quad \text{Surv}(\text{EWBG}_{\text{SM}}) = 0.$$

Those conclusions occur only after diagnostic admission, verdict-map construction, no-fifth-function exhaustion, the imported physical-Higgs crossover theorem, and the imported measured minimal-SM EWBG failure theorem are applied.

Proof. The definitions of the targets fix the carrier inventory, measured anchors, same-scope transformations, and candidate route spaces. They do not delete C_{lower} , select C_{cross} , delete C_{1st} , or delete EWBG_{SM} . A lower branch is admitted as diagnostic material before route exhaustion. Crossover is imported only as a physical-Higgs finite-temperature theorem. Minimal-SM EWBG failure is imported only after the successful-route target is defined. Therefore no conclusion is smuggled into the target. $\square$

3.1 Typed target split

Define $D_{\text{HV},0}^{T=0}$ as the zero-temperature stability and lifetime target, and $D_{\text{HV},0}^{T>0}$ as the finite-temperature transition target. A zero-temperature false-vacuum claim is not a thermal transition-order claim. A finite-temperature crossover theorem is not an absolute-stability theorem. This target split is a standing requirement, not a stylistic preference.

4 Endpoint status calculus

Definition 4.1 (Availability, standing, survivor status, and selection). For fixed target D and candidate class C :

- $\text{Avail}_D(C) = 1$ means C is formally writable or diagnostically available in the declared arena;
- $\text{Stand}_D(C) = 1$ means C has endpoint-usable status after quotient/admissibility filters;
- $\text{Surv}_D(C) = 1$ means C is not deleted, quotient-collapsed, or moved to richer scope;
- $\text{Sel}_D(C) = 1$ means C is the unique certified survivor quotient of the target.

Availability is not occupancy. Survival is not selection. Richer-scope update is not same-scope survival.

Criterion 4.2 (Standing-preserving endpoint selection). For a fixed target D with quotient relation $\sim_D$,

$$\text{Sel}_D(C) = 1 \iff \text{Avail}_D(C) = 1 \wedge \text{Stand}_D(C) = 1 \wedge \forall C' \not\sim_D C, \text{Surv}_D^{\text{cert}}(C') = 0.$$

Selection is uniqueness over certified endpoint-status quotients, not over written representatives.

Definition 4.3 (Endpoint, diagnostic, and route classes). For $D_{\text{HV},0}^{T=0}$ the physical endpoint classes are C_{abs} , C_{meta} , and C_{unst} . The candidate primitive lower-vacuum class C_{lower} is the claim that a lower branch of a Higgs effective-potential representative itself carries primitive same-scope endpoint standing in the measured Standard Model target. Its standing is not decided by definition; the exhaustion theorems below determine whether such a primitive lower-branch role survives. Precision sensitivity is not an endpoint class. It is an audit verdict:

$$\text{Audit}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(D_{\text{HV},0}) = \text{PrecisionSensitive}.$$

For $D_{\text{HV},0}^{T>0}$ the same-scope endpoint classes are C_{cross} , C_{1st} , and C_{2nd} . Richer-scope thermal behavior is not a same-scope endpoint class; it is a target-update status in $D_{\text{HV},+}$. For $D_{\text{EWBG},0}$, the target route is EWBG_{SM} , the measured minimal-Standard-Model route that would produce and preserve the observed baryon asymmetry by electroweak dynamics alone.

5 Local AASC proof calculus

The proof uses a local AASC calculus stated here so that the paper does not rely on unstated source-corpus conventions. A *target* fixes the object of evaluation, the carrier inventory, the admitted representatives, and the transformations counted as same-scope. A *representative* is a calculational or coordinate presentation inside a target. A *diagnostic* is a representative feature usable in calculations but not itself a selected physical endpoint. A *primitive endpoint* is a claim used as endpoint standing without first being derived from a quotient-stable stability, lifetime, or richer-scope certificate. An *endpoint class* is a quotient-stable verdict class. A *richer-scope update* adds carrier, law, threshold, boundary, witness, or UV structure and therefore changes the target.

Criterion 5.1 (No-promotion). *A representative, diagnostic, coordinate branch, boundary convention, or measured anchor has no primitive endpoint standing merely because it is formally writable or useful in a calculation. It becomes endpoint-bearing only after it descends to a quotient-stable endpoint class or is declared as a target update.*

Criterion 5.2 (Exhaustive classification as proof). *For a fixed target, once the candidate classes, admissible transformations, verdict maps, quotient relations, and endpoint-selection criterion are fixed, deleting every inequivalent certified survivor except one proves selection of the remaining class. Conversely, if an inequivalent rival cannot be deleted under the declared uncertainty and transport envelope, no singleton selection follows.*

Lemma 5.3 (Rival-basin non-deletion). *Let an anchor and uncertainty envelope U intersect two inequivalent endpoint basins B_1 and B_2 for a verdict map ν . If no declared certificate separates U from one of the basins by the required threshold Δ , then neither associated endpoint class is deleted by the endpoint-selection criterion.*

Proof. Selection requires certified deletion of every inequivalent rival. If $U \cap B_1 \neq \emptyset$ and $U \cap B_2 \neq \emptyset$, then both verdicts remain compatible with the admitted source-state envelope. Deleting either one would require an additional separation, matching, perturbative, statistical, or high-scale certificate excluding that basin. Without such a certificate the rival remains a certified live basin, so singleton selection is blocked. $\square$

6 Kernel necessity for Higgs-vacuum endpoint claims

Theorem 6.1 (Higgs-vacuum endpoint use is governed construction). *For any candidate class C used as a claim-bearing endpoint of $D_{\text{HV},0}$,*

$$\text{EndpointUse}_{D_{\text{HV},0}}(C) \Rightarrow K_D = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}).$$

Consequently Higgs-vacuum endpoint claims are subject to target fixation, quotient construction, gauge-vacuum compatibility, RG/scheme transport discipline, no-promotion, and richer-scope update accounting.

Proof. A claim-bearing Higgs-vacuum endpoint must specify which object is evaluated: a zero-temperature vacuum fate, a decay/lifetime class, a finite-temperature transition class, or a richer-scope completion. That is reference. It must distinguish a raw expression, a diagnostic, a decay rate, a thermodynamic class, an empirical anchor, and a target update. That is standing. It must identify which gauge, scheme, RG, matching, and field-redescription moves preserve the same claim and which moves alter the target. That is admissibility. It must prevent a failed same-scope branch from being repaired afterward by hidden new physics while still being counted as the same endpoint act. That is irreversibility. These roles are already required by determinate endpoint use. $\square$

Corollary 6.2 (Dependency-inversion exclusion). *A displayed effective-potential branch, an RG crossing, a chosen gauge representative, or a preferred top-mass input cannot define the standing rule by which it becomes endpoint-selected.*

7 Explicit quotient construction

7.1 Admissible transformations

Let $\mathcal{T}_0^{\text{stat}}$, $\mathcal{T}_0^{\text{life}}$, and $\mathcal{T}_T^{\text{therm}}$ denote the admissible transformation families for the three typed targets. They include, as appropriate:

- gauge-fixing changes controlled by Nielsen/Fukuda-Kugo identities;
- renormalization-scale and renormalization-scheme changes;
- threshold and matching convention changes;
- field-coordinate redefinitions preserving the carrier and physical witness content;
- top-mass and coupling conversion maps between measured anchors and Lagrangian/RG inputs;
- perturbative-order changes accompanied by a declared uncertainty model;
- thermal dimensional-reduction and resummation maps preserving the thermodynamic class.

A transformation that adds a new scalar, symmetry, threshold, UV boundary condition, or gravitational rule is not an admissible same-scope transformation. It is a map to $D_{\text{HV},+}$.

7.2 Verdict maps before quotienting

The quotient relations are not defined by an unexplained verbal appeal to “same verdict.” The verdict maps are part of the construction. Let

$$\begin{aligned}\nu_{\text{stat}} : \mathcal{R}_{T=0}^{\text{stat}} &\longrightarrow \{\text{NoDeeperBranch}, \text{LowerBranchDiagnostic}, \text{UncontrolledStatic}\}, \\ \nu_{\text{life}} : \mathcal{R}_{T=0}^{\text{life}} &\longrightarrow \{\text{EffectivelyStable}, \text{Metastable}, \text{Unstable}, \text{UncontrolledLifetime}\}, \\ \nu_{\text{therm}} : \mathcal{R}_{T>0}^{\text{therm}} &\longrightarrow \{\text{Crossover}, \text{FirstOrder}, \text{Critical}, \text{UncontrolledThermal}\}.\end{aligned}$$

The domains are respectively static effective-potential diagnostic presentations, zero-temperature decay/lifetime presentations, and finite-temperature thermodynamic presentations admitted by the fixed target. Each verdict map is partial until the required gauge, transport, matching, perturbative-order, anchor, and uncertainty controls are declared. When those controls are missing, the representative is assigned the relevant uncontrolled audit value rather than a physical endpoint.

The uncontrolled values in the verdict-map codomains are audit residues. They define equivalence among non-selectable presentations for purposes of route closure, reproducibility, and rival-basin accounting, but they do not define endpoint classes eligible for Sel. Selection requires a physical verdict value—such as effective stability, metastability, instability, crossover, first-order transition, or criticality—or a richer-scope update certificate. Thus an uncontrolled quotient can preserve a record of unresolved diagnostic status without becoming an endpoint.

The lifetime verdict map is operational. A lifetime presentation L contains at least a decay exponent or bounce-action estimate B , a prefactor or bounded prefactor envelope A , a decay probability density per spacetime volume

$$\Gamma_4 \sim Ae^{-B},$$

a declared comparison four-volume $\mathcal{V}_4$, and an uncertainty envelope for gauge, scheme, matching, and perturbative inputs. The decay probability over that comparison four-volume is represented schematically by

$$P_{\text{dec}}(\mathcal{V}_4) = 1 - \exp[-\Gamma_4 \mathcal{V}_4].$$

The verdict $\nu_{\text{life}}(L)$ is Metastable only when a lower branch is admitted and the decay probability is below the declared comparison threshold; it is Unstable when the threshold is exceeded; it is EffectivelyStable when no lower decay channel survives the static and lifetime quotients; and it is UncontrolledLifetime when the required lifetime data or uncertainty propagation is not closed. This is the Coleman-Callan false-vacuum logic expressed as a status map rather than as a new decay calculation [11, 12].

Definition 7.1 (Lifetime-certificate component). The lifetime component of a zero-temperature singleton certificate is

$$\mathcal{L} = (\mathcal{V}_4, \epsilon, \Pi_{\Gamma_4}, \mathcal{E}_{\text{life}}, \mathcal{U}_{\text{grav}}),$$

where $\mathcal{V}_4$ is the declared comparison four-volume, ϵ is the admissible decay-probability threshold, Π_{Γ_4} is the decay-rate prescription including bounce action, prefactor treatment, and renormalization prescription, $\mathcal{E}_{\text{life}}$ is the propagated gauge/scheme/matching/perturbative uncertainty envelope, and $\mathcal{U}_{\text{grav}}$ declares whether gravitational or high-scale corrections are excluded, included, or target-updating. The lifetime certificate uses $\mathcal{L}$ as a certificate component; a new decay-rate computation is theorem-relevant here only if it supplies, breaks, or updates Π_{Γ_4} , ν_{life} , or the declared source-state ledger.

Definition 7.2 (Admissibility-relevant invariant bundle). For a fixed Higgs-vacuum target D , let I_D denote the admissibility-relevant invariant bundle: the family of declared invariants sufficient to determine target identity, same-scope admissibility, reuse-stable standing, and the boundary between licensed continuation and target-changing repair. A condition belongs to I_D when varying it would leave uncertain whether two constructions evaluate the same target, carry the same standing-bearing status, or preserve the same admissibility boundary.

Lemma 7.3 (High-scale continuation as invariant-bundle content). *For the zero-temperature measured minimal-SM lifetime target $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$, the high-scale continuation clause*

$$\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$$

belongs to I_D . Here $\Lambda_{\mathcal{P}}$ is the declared protocol/support-envelope scale for the submitted lifetime calculation: it may be chosen to dominate the field, running, and bounce support required by the protocol, but it is declared before the endpoint verdict is treated as standing-bearing and is not selected afterward as a repair of the construction. The clause $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ states that the Standard Model carrier inventory, matching prescription, and RG transport rules remain the declared comparison domain up to that support envelope, with no intervening threshold, carrier update, source-rule change, or target-changing high-scale boundary below it.

Proof. Reference requires $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$. Without it, a high-field calculation is not tied to a determinate target: one representative may evaluate continued minimal-SM transport while another evaluates a threshold-modified, gravity-modified, or EFT-broken target, even though both use low-energy anchors. The continuation clause fixes what counts as the same target under high-scale redescription.

Admissibility also requires $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$. The gate must distinguish same-domain RG and lifetime continuation from an update to $D_{\text{HV},+}$. If the continuation rule is implicit or variable, the

boundary between licensed minimal-SM extrapolation and target-changing high-scale modification is not jointly enforceable. Declaring $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ turns that boundary into a binary condition: a construction either respects the invariant bundle or exits to $D_{\text{HV},+}$.

Standing requires the same clause. A verdict such as $\nu_{\text{life}}(B) = \text{Metastable}$ is reuse-stable only under the invariants used to compute it. If the high-scale continuation rule could be altered later while still treating the result as the same endpoint act, later target change would function as retrospective repair. Irreversibility forbids that move. A change in $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$, or a post-hoc enlargement of $\Lambda_{\mathcal{P}}$ after the verdict is asserted, is therefore a new construction on an updated target, not a repair of the original construction. Since gate entry is binary, graded judgments about how trustworthy the extrapolation is may appear downstream inside uncertainty analysis, but they do not replace the AMetric boundary condition itself. Hence $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}}) \in I_D$. $\square$

Corollary 7.4 (No-new-physics continuation is not primitive standing). *A no-new-physics high-scale continuation assumption does not add particles, forces, or BSM operators in the model-building sense. Nevertheless it is not dispensable for endpoint standing. It is invariant-bundle content required for a same-domain lifetime verdict:*

$$\text{NoBSM}(\Lambda_{\mathcal{P}}) \not\equiv \text{Stand}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}),$$

while a closed lifetime route can certify metastability:

$$\text{NoBSM}(\Lambda_{\mathcal{P}}) + I_D + \mathcal{S}_{\text{HV}}^{T=0} + \mathcal{L} + \nu_{\text{life}}(B) = \text{Metastable} \quad \Rightarrow \quad C_{\text{meta}}$$

only when the invariant bundle I_D , the zero-temperature singleton certificate $\mathcal{S}_{\text{HV}}^{T=0}$, and the lifetime package $\mathcal{L}$ close under the declared separation threshold.

Lemma 7.5 (Compatibility with standard minimal-SM metastability calculations). *A calculation that fixes the measured minimal-SM anchor package, specifies the top-mass conversion and matching prescription, performs RG transport in a declared scheme, constructs a gauge-controlled effective-potential diagnostic, declares $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ as part of the protocol support envelope, and computes a bounce/lifetime verdict under a declared comparison four-volume and uncertainty model is an admissible route through ν_{life} . If that route closes with a long-lived metastability verdict and the singleton-certificate components are closed above the declared separation threshold, the calculation certifies C_{meta} on the fixed minimal-SM domain. It does not certify primitive endpoint standing for C_{lower} , because the endpoint object is the lifetime verdict under I_D , not the raw lower branch of an effective-potential representative.*

Proof. Each component of the standard calculation supplies governance work required by the lifetime verdict map. The measured anchors fix the low-energy input ledger; top-mass conversion and matching fix the field-theoretic input used by the running Yukawa and quartic couplings; RG transport fixes the same-scope high-scale representative family; gauge-controlled effective-potential extraction prevents representative field-coordinate structure from being read as endpoint standing; $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ fixes the high-scale comparison domain; and the bounce/lifetime package supplies Γ_4 , B , $\mathcal{V}_4$, ϵ , and the propagated uncertainty envelope. These data are exactly the conditions under which ν_{life} can assign a physical lifetime verdict. A long-lived verdict therefore licenses C_{meta} if the singleton certificate closes. The raw lower branch remains diagnostic material: its endpoint-bearing content has been mediated by ν_{life} and I_D , so it is not primitive standing for C_{lower} . $\square$

7.3 Exhaustion of same-domain route availability

Definition 7.6 (Candidate route protocol). A candidate route protocol for C_{meta} in a fixed target D is a well-typed construction protocol $\mathcal{P}$ that purports to make C_{meta} route-available by entering a verdict map, without yet asserting that the route is available or that the endpoint has standing. We write

$$\text{CandRoute}_D(C_{\text{meta}}; \mathcal{P})$$

for this candidate status. Candidate status is submitted to route-status classification; it is not itself route availability and carries no endpoint standing.

Definition 7.7 (Same-domain route invariants). A candidate protocol $\mathcal{P}$ remains same-domain for C_{meta} only if it preserves the route invariants of the fixed target $D_{\text{HV},0}^{T=0}[\mathbf{S}_{2026}]$:

$$(A_{\text{HV}}, I_D, \text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}}), \nu_{\text{life}}, \mathcal{L}, \mathbf{S}).$$

Thus the measured anchor package, admissibility-relevant invariant bundle, declared high-scale continuation clause, lifetime verdict map, lifetime package, and source-state discipline are fixed before route status is assessed. Any alteration of the carrier inventory, threshold structure, high-scale continuation clause, verdict map, source state, or admissibility boundary is a target update rather than a same-domain route.

Definition 7.8 (Lifetime-route normal form and partial normalizer). The lifetime-route normal form for same-domain metastability availability is the tuple

$$\mathcal{R}_{\text{life}} = (A_{\text{HV}}, I_D, \text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}}), T_{\text{RG}}, B, \mathcal{L}, U_{\Delta}, \mathbf{S}),$$

where A_{HV} is the measured minimal-SM anchor package; I_D is the admissibility-relevant invariant bundle; $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ is the declared high-scale continuation over the protocol support envelope; T_{RG} is the RG, scheme, matching, and top-mass transport package; B is the gauge-controlled effective-potential branch diagnostic; $\mathcal{L}$ is the bounce/lifetime verdict package; U_{Δ} is the uncertainty and rival-basin separation envelope; and $\mathbf{S}$ is the source state in which the route is evaluated.

The partial normalizer $\text{NF}_{\text{life}}(\mathcal{P})$ is defined, written $\text{NF}_{\text{life}}(\mathcal{P}) \downarrow$, exactly when $\mathcal{P}$ preserves the fixed target and supplies all components of the lifetime-route normal form. When defined,

$$\text{NF}_{\text{life}}(\mathcal{P}) = \mathcal{R}_{\text{life}}$$

with the components instantiated by the declared protocol data of $\mathcal{P}$. For diagnostic-only, route-incomplete, target-updating, or standing-smuggling protocols, $\text{NF}_{\text{life}}(\mathcal{P})$ is undefined.

Definition 7.9 (Route-form equivalence). For fixed target D , write

$$\mathcal{P} \cong_D \mathcal{P}'$$

iff $\mathcal{P}$ and $\mathcal{P}'$ preserve the same fixed target D , admissibility-relevant invariant bundle I_D , declared high-scale continuation clause $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$, verdict-map entry through ν_{life} , lifetime package $\mathcal{L}$, uncertainty/separation envelope U_{Δ} , and source-state discipline, up to notation, scheme presentation, or representative bookkeeping that does not alter route status. Route-form equivalence is therefore weaker than syntactic identity but stronger than informal pipeline similarity: it preserves exactly the route-form data that can bear same-domain availability for C_{meta} .

Definition 7.10 (Ordered fail-closed route-status classifier). For a candidate metastability route $\mathcal{P}$, the route-status classifier is assigned by first applicable fail-closed status in the priority order

TargetUpdate $\succ$ StandingSmuggling $\succ$ DiagnosticOnly $\succ$ RouteIncomplete $\succ$ RouteNormal.

The classifier is written $\rho_{\text{route}}^D(\mathcal{P})$. Its clauses are:

1. TargetUpdate: $\mathcal{P}$ changes I_D , $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$, the source state, carrier inventory, threshold structure, admissibility boundary, or verdict map.
2. StandingSmuggling: $\mathcal{P}$ treats diagnostic availability, technical success, route availability, or later precision as endpoint standing before certificate closure.
3. DiagnosticOnly: $\mathcal{P}$ supplies branch diagnostic content but no ν_{life} -ready route.
4. RouteIncomplete: $\mathcal{P}$ preserves the fixed target but lacks at least one forced component of $\mathcal{R}_{\text{life}}$.
5. RouteNormal: $\mathcal{P}$ preserves the fixed target and $\text{NF}_{\text{life}}(\mathcal{P}) \downarrow$.

The priority order makes ρ_{route}^D functional even when informal descriptions overlap. For example, a target-changing incomplete protocol is classified as TargetUpdate, and a route-normal-looking protocol that treats availability as standing closure is classified as StandingSmuggling.

Definition 7.11 (Route availability after classification). Route availability is reserved for candidate protocols that survive the ordered classifier as route-normal:

$$\text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1 \iff \text{CandRoute}_D(C_{\text{meta}}; \mathcal{P}) \wedge \rho_{\text{route}}^D(\mathcal{P}) = \text{RouteNormal}.$$

Let

$$\mathfrak{R}_D(C_{\text{meta}}) := \{\mathcal{P} : \text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1\}$$

be the class of same-domain protocols with route availability after classification. The shorthand

$$\text{RouteAvail}_D(C_{\text{meta}}) = \text{RouteAvail}_{\mathcal{R}_{\text{life}}}(C_{\text{meta}})$$

means the typed class statement that $\mathfrak{R}_D(C_{\text{meta}})$ consists exactly of route-normal protocols whose partial lifetime normalizer is defined, up to $\cong_D$. It is a statement about route-form equivalence classes, not endpoint standing.

Route availability is not standing closure:

$$\text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1 \not\Rightarrow \text{StandingClosed}_D(C_{\text{meta}}; \mathcal{P}) = 1$$

is not a valid inference without the closed singleton and lifetime certificates.

Lemma 7.12 (Forced components of a ν_{life} -ready route). *A same-domain ν_{life} -ready route to C_{meta} requires every component of*

$$\mathcal{R}_{\text{life}} = (A_{\text{HV}}, I_D, \text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}}), T_{\text{RG}}, B, \mathcal{L}, U_{\Delta}, \mathbf{S}).$$

Proof. The anchor package A_{HV} fixes measured-target reference. The invariant bundle I_D fixes same-target and same-boundary invariance. The high-scale continuation clause $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ fixes the support envelope on which the minimal-SM route remains same-domain. The transport package T_{RG} connects the measured anchors to the high-scale representative used by the lifetime calculation. The diagnostic branch B supplies the lower channel under gauge-controlled effective-potential extraction. The lifetime package $\mathcal{L}$ supplies the bounce, decay-rate, comparison-volume, and threshold data needed by ν_{life} . The envelope U_{Δ} records uncertainty and rival-basin separation. The source state $\mathbf{S}$ fixes which anchors, uncertainties, and source commitments are being evaluated. Omitting any component either removes the lifetime verdict, leaves the target indeterminate, or changes the source/target domain; hence a same-domain ν_{life} -ready route requires the entire normal form. $\square$

Lemma 7.13 (Diagnostic branch alone is route-incomplete). *A lower branch B of a same-scope effective-potential representative gives diagnostic availability,*

$$B \Rightarrow \text{DiagnosticAvailability} = 1,$$

but it does not by itself give same-domain route availability for C_{meta} . Unless B is embedded into a candidate whose partial lifetime normalizer is defined, $\rho_{\text{route}}^D(\mathcal{P})$ is DiagnosticOnly or RouteIncomplete, subject to the priority order of Definition 7.10.

Proof. The branch diagnostic supplies neither the declared high-scale continuation clause, nor the RG/matching/top-mass transport bundle, nor the bounce/lifetime package, nor the uncertainty and rival-basin separation envelope. It therefore cannot enter ν_{life} as a route-normal object by itself. It is admitted as diagnostic content, but not as route availability or endpoint standing. $\square$

Lemma 7.14 (No non-lifetime same-domain route to metastability). *Within $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$, a same-domain route to C_{meta} must enter through ν_{life} . It cannot be supplied by ν_{stat} , ν_{therm} , or raw diagnostic availability.*

Proof. Metastability is a false-vacuum lifetime status: it asserts an admitted lower channel whose decay probability remains below the declared comparison threshold. Static ordering can diagnose whether a lower channel is present or absent, but not whether the electroweak vacuum is long-lived on the declared comparison domain. Finite-temperature transition structure belongs to $D_{\text{HV},0}^{T>0}$, not to the zero-temperature lifetime target. Raw diagnostic availability supplies no lifetime verdict. Therefore the only same-domain verdict map capable of licensing C_{meta} is ν_{life} . $\square$

Lemma 7.15 (Continuation variation exits same-domain route availability). *If a candidate route alters $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ after protocol declaration, or varies it while claiming same-domain route status, then it does not preserve the same I_D . It is a target update, a new source-state construction, or an ill-typed repair attempt.*

Proof. By Lemma 7.3, $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}}) \in I_D$. Varying it changes the high-scale comparison domain over which RG transport, branch diagnostics, and lifetime verdicts are evaluated. If the variation is declared prospectively, it defines an updated target or source state. If it is introduced after the endpoint verdict to preserve the original claim, it is a same-act repair. Irreversibility blocks the latter, and target fixation classifies the former as $D_{\text{HV},+}$ or a new source-state construction. $\square$

Lemma 7.16 (Pipeline availability is not standing transfer). *Technical success of a route pipeline does not transfer standing from diagnostic availability to endpoint standing:*

$$\text{PipelineSuccess}(\mathcal{P}) \not\Rightarrow \text{StandingClosed}_D(C_{\text{meta}}; \mathcal{P}).$$

Proof. A technical pipeline can supply components of a candidate route, but standing is not generated by technical success, transferred by a pipeline, or repaired after an incomplete act. The no-generator/no-carrier/no-repair impossibility suite records precisely those prohibitions: legitimacy is produced only by the gate, does not transfer across domains or pipelines, and cannot be recovered post hoc [32]. Therefore a pipeline contributes to standing only when its candidate protocol passes the route classifier and the singleton/lifetime certificate closes. $\square$

Lemma 7.17 (Route incompleteness is not a third endpoint status). *A protocol with $\rho_{\text{route}}^D(\mathcal{P}) = \text{RouteIncomplete}$ does not create an intermediate endpoint status. It is not standing-bearing in the current source state, although a later source state may supply the missing components prospectively.*

Proof. Endpoint selection is over certified survivors. A route-incomplete protocol lacks at least one component required for $\mathcal{R}_{\text{life}}$ or the singleton certificate. Treating that incompleteness as a third endpoint would introduce a non-bivalent middle standing status. The fixed-domain exhaustion framework rejects such a move fail-closed: admissibility-bearing variation must enter through the closed local loci of the fixed comparison class, not through a new undefined midpoint [31]. $\square$

Lemma 7.18 (Route-status classifier completeness). *Every candidate C_{meta} -route protocol in the fixed zero-temperature target receives exactly one route status by the ordered fail-closed classifier:*

$$\text{CandRoute}_D(C_{\text{meta}}; \mathcal{P}) \Rightarrow \rho_{\text{route}}^D(\mathcal{P}) \in \left\{ \begin{array}{l} \text{DiagnosticOnly, RouteIncomplete,} \\ \text{RouteNormal, TargetUpdate,} \\ \text{StandingSmuggling} \end{array} \right\}.$$

No sixth same-domain route-availability class remains.

Proof. The priority order makes the classifier functional. If $\mathcal{P}$ changes the fixed target, invariant bundle, source state, carrier inventory, threshold structure, admissibility boundary, or verdict map, it is classified first as TargetUpdate. If it preserves those items but treats diagnostic availability, technical success, route availability, or later precision as endpoint standing before certificate closure, it is StandingSmuggling. If it supplies only lower-branch diagnostic content, it is DiagnosticOnly. If it preserves the fixed target but lacks at least one forced component of Lemma 7.12, it is RouteIncomplete. If none of those failure classifications applies, then the target is preserved and every forced component is supplied, so $\text{NF}_{\text{life}}(\mathcal{P}) \downarrow$ and $\mathcal{P}$ is RouteNormal. These cases exhaust the possible governance-bearing locations for a candidate route: target change, illicit standing conversion, diagnostic content, incomplete normal-form content, or complete normal-form content. A sixth class would have to produce same-domain governance-bearing variation while leaving boundary typing, standing partition, lawful quotient, and admissible continuation unchanged, which is excluded by fixed-domain exhaustion [31]. $\square$

Theorem 7.19 (Exhaustion of same-domain metastability route availability). *Within the fixed zero-temperature measured minimal-SM Higgs-vacuum target $D = D_{\text{HV},0}^{T=0}[\mathcal{S}]$, every candidate route protocol for C_{meta} is classified by the ordered fail-closed route-status classifier:*

$$\text{CandRoute}_D(C_{\text{meta}}; \mathcal{P}) \Rightarrow \rho_{\text{route}}^D(\mathcal{P}) \in \left\{ \begin{array}{l} \text{DiagnosticOnly, RouteIncomplete,} \\ \text{RouteNormal, TargetUpdate,} \\ \text{StandingSmuggling} \end{array} \right\}.$$

Moreover, route availability survives only in the route-normal case:

$$\begin{aligned} \text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1 &\Rightarrow \text{NF}_{\text{life}}(\mathcal{P}) \downarrow \\ &\wedge \mathcal{P} \cong_D \text{NF}_{\text{life}}(\mathcal{P}). \end{aligned}$$

Equivalently,

$$\mathfrak{R}_D(C_{\text{meta}})/\cong_D = \{\mathcal{P} : \text{CandRoute}_D(C_{\text{meta}}; \mathcal{P}) \wedge \rho_{\text{route}}^D(\mathcal{P}) = \text{RouteNormal}\}/\cong_D.$$

This is route-form equality, not standing closure.

Proof. Let $\mathcal{P}$ be a candidate route protocol for C_{meta} . Lemma 7.18 gives exactly one fail-closed route status. If the status is TargetUpdate, the protocol has left the fixed target. If it is StandingSmuggling, the protocol tries to convert technical or route availability into endpoint standing without the gate.

If it is DiagnosticOnly, the protocol supplies only branch material. If it is RouteIncomplete, it lacks at least one forced component of $\mathcal{R}_{\text{life}}$. None of those cases yields $\text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1$, because route availability is reserved by definition for RouteNormal.

If $\text{RouteAvail}_D(C_{\text{meta}}; \mathcal{P}) = 1$, then $\rho_{\text{route}}^D(\mathcal{P}) = \text{RouteNormal}$. By the classifier definition and Lemma 7.12, the fixed target is preserved and all forced components of the lifetime-route normal form are supplied. Hence $\text{NF}_{\text{life}}(\mathcal{P}) \downarrow$. Since those components determine the same target, invariant bundle, high-scale continuation clause, lifetime map, lifetime package, uncertainty/separation envelope, and source-state discipline, $\mathcal{P} \cong_D \text{NF}_{\text{life}}(\mathcal{P})$. Thus every route-available protocol factors through the partial lifetime normalizer.

The exhaustion is structural rather than enumerative. The fixed-domain exhaustion theorem reduces same-scope governance-bearing variation to the closed local loci of boundary typing, standing partition, lawful quotient, and admissible continuation, with no fifth locus remaining [31]. The lifetime-route normal form is the Higgs specialization of those loci for C_{meta} . The no-generator/no-carrier/no-repair impossibility package blocks technical route success from becoming an independent authorizer, transfer channel, or post-hoc repair [32]. Therefore no same-domain route-availability class remains outside the route-normal normalizer. $\square$

Corollary 7.20 (Route availability does not revive primitive lower-branch standing). *If $\text{RouteAvail}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) = 1$, then $\mathcal{P}$ is a route-normal lifetime protocol, but*

$$\text{RouteAvail}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) = 1 \not\Rightarrow \text{Stand}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}).$$

The endpoint object of a route-normal metastability construction is the lifetime verdict C_{meta} , not the representative lower branch.

Corollary 7.21 (Current-source route availability remains precision-sensitive). *A source-state route may be available in normal form,*

$$\text{RouteAvail}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) = 1,$$

while still failing to select C_{meta} if U_{Δ} , $\mathcal{L}$, or $\mathcal{S}_{\text{HV}}^{T=0}$ does not close. Thus route availability in $\mathcal{S}_{2026}$ is compatible with the audit verdict

$$\text{Audit}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(D_{\text{HV},0}) = \text{PrecisionSensitive}.$$

Corollary 7.22 (Future route closure selects metastability, not primitive branch standing). *If a later source state $\mathcal{S}_+$ closes I_D , $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$, $\mathcal{L}$, U_{Δ} , and $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}})$, then the selected endpoint is C_{meta} , not C_{lower} with primitive standing.*

Table 1: Same-domain route availability classification for C_{meta} .

Proposed route claim	Classification	Result
<i>Classification is by first applicable fail-closed status in the priority order $\text{TargetUpdate} \succ \text{StandingSmuggling} \succ \text{DiagnosticOnly} \succ \text{RouteIncomplete} \succ \text{RouteNormal}$.</i>		
Lower branch exists in V_{eff}	Diagnostic-only	Admitted diagnostic; no route standing.

RG-improved potential only	Route-incomplete	Needs lifetime package and uncertainty/separation envelope.
RG plus bounce without declared $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$	Route-incomplete or target-indeterminate	Same-domain target is not fixed.
RG plus bounce plus declared continuation and uncertainty envelope	Route-normal	Admissible ν_{life} route; standing still requires certificate closure.
BSM threshold, Planck operator, or changed high-scale rule	Target update	Moves to $D_{\text{HV},+}$.
Route exists, therefore C_{meta} stands	Standing-smuggling	Blocked by no-generator, no-carrier, and no-repair discipline.
Later precision repairs an earlier incomplete route	Standing-smuggling or new source state	Forward update only, not retroactive repair.

Theorem 7.23 (Verdict-map completeness for lower-branch endpoint function). *Let B be a lower branch of a Higgs effective-potential representative admitted as same-scope diagnostic material in $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$. Any standing-bearing claim that B performs endpoint work must be recorded by one of the typed verdict maps or by a target update:*

$$\nu_{\text{stat}}(B_R), \quad \nu_{\text{life}}(B_L), \quad \nu_{\text{therm}}(B_F), \quad D_{\text{HV},0} \rightarrow D_{\text{HV},+},$$

where B_R is the static representative, B_L is a lifetime/decay presentation when a lower channel is admitted, and B_F is a finite-temperature presentation when transition order is claimed. If none of these maps assigns a physical verdict value and no target update is supplied, the branch has only diagnostic availability and cannot be selected as an endpoint.

Proof. The verdict maps were introduced before the quotient relations precisely to fix the possible endpoint functions. In the zero-temperature static ledger, a lower branch can affect only whether the electroweak vacuum has no deeper same-scope competitor, has a lower-branch diagnostic, or remains uncontrolled; that is ν_{stat} . If the branch is used to claim false-vacuum fate, the relevant object is no longer the plotted branch but the decay/lifetime presentation with Γ_4 , B , $\mathcal{V}_4$, and an uncertainty envelope; that is ν_{life} . If the branch is used to claim transition order, the target is finite-temperature thermodynamics and the relevant map is ν_{therm} . If the branch is made endpoint-bearing only by adding a scalar, threshold, symmetry, gravitational boundary, UV condition, or other carrier/source, the claim has left the fixed target and becomes $D_{\text{HV},+}$. These cases exhaust the ways a lower branch can change endpoint status while preserving target typing. A remaining candidate supplies neither a static verdict, nor a lifetime verdict, nor a thermal verdict, nor a carrier update; it is therefore a representative diagnostic, not endpoint standing. $\square$

Theorem 7.24 (No fifth same-scope lower-branch endpoint function). *Let B be a lower branch of an admitted Higgs effective-potential representative. If B is used as endpoint-bearing content in the fixed measured Higgs target, then the asserted endpoint function of B is one of four and only four types: static vacuum ordering or stability classification, governed by ν_{stat} ; false-vacuum decay or lifetime classification, governed by ν_{life} ; finite-temperature transition classification, governed by ν_{therm} ; or richer-scope source, carrier, boundary, or UV update, governed by $D_{\text{HV},0} \rightarrow D_{\text{HV},+}$. Any proposed fifth use is either diagnostic bookkeeping, reduces to one of these functions, or changes target.*

Proof. A lower branch can perform endpoint work only by changing a claim-bearing ledger. An endpoint-bearing use of a lower branch must either change the zero-temperature ordering of admissible vacua, change the decay probability of the electroweak vacuum, change the finite-temperature transition history, or introduce a carrier/source condition not present in the fixed target. These alternatives exhaust the possible ways in which a lower branch can alter the status of the fixed Higgs-vacuum target: every endpoint-bearing claim must change a zero-temperature status, a finite-temperature status, or the target inventory; anything outside those alternatives is diagnostic bookkeeping rather than endpoint work. If it compares vacuum depth or the presence of a deeper same-scope channel, it is a static-ordering input and factors through ν_{stat} . If it supports a claim that the electroweak vacuum decays or does not decay on a declared comparison domain, it is a lifetime input and factors through ν_{life} . If it is used to classify thermal history, bubble nucleation, latent heat, or transition order, it belongs to the finite-temperature target and factors through ν_{therm} . If its endpoint force depends on a new scalar, threshold, symmetry, gravitational boundary, cosmological history, UV completion, or other source-continuity carrier, the original target has been updated to $D_{\text{HV},+}$. A remaining use changes none of the endpoint ledgers and therefore remains a representative diagnostic. Since the lower branch is a candidate endpoint only by affecting a status of the fixed Higgs-vacuum target, and the target has only zero-temperature, finite-temperature, and inventory/update status ledgers, no further same-scope endpoint function remains. A proposed fifth endpoint function has standing only if it identifies a fifth target-status ledger while preserving the same target. $\square$

Criterion 7.25 (Standing test for a fifth endpoint function). *A same-target objection asserting a fifth lower-branch endpoint function must supply a map $F_5(B)$ with standing. The standing condition is:*

$$\text{Stand}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(F_5(B)) = 1$$

only if all of the following hold:

$$F_5(B) \not\equiv \nu_{\text{stat}}(B_R), \quad F_5(B) \not\equiv \nu_{\text{life}}(B_L), \quad F_5(B) \not\equiv \nu_{\text{therm}}(B_F), \quad F_5(B) \not\equiv D_{\text{HV},+},$$

while $F_5(B)$ preserves the same fixed target $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$ and imports no new carrier, source, boundary, witness, repair load, thermal target, or cosmological target. A proposed $F_5(B)$ that fails any condition is diagnostic bookkeeping, verdict-map content, or target update rather than a fifth same-scope endpoint function.

Corollary 7.26 (Fifth-function closure entails primitive lower-branch deletion). *If the verdict maps and target-update route are exhausted and no $F_5(B)$ satisfying Criterion 7.25 is supplied, then the primitive lower-branch claim-form has no same-scope survivor:*

$$\neg \exists F_5(B) \wedge \text{Exhaust}(\nu_{\text{stat}}, \nu_{\text{life}}, \nu_{\text{therm}}, D_{\text{HV},+}) \Rightarrow \text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0.$$

This corollary is the direct bridge from no-fifth-function closure to primitive lower-branch deletion. It does not deny certificate-dependent metastability; it denies only the primitive claim that the lower branch itself has endpoint standing independently of the verdict maps or a target update.

Table 2: Reduction of alleged fifth lower-branch endpoint functions.

Alleged fifth function	AASC reduction
RG instability scale	Static or lifetime diagnostic; not an endpoint without ν_{stat} or ν_{life} .
Lower field-coordinate minimum	Gauge/scheme representative; no endpoint standing without static or lifetime quotient descent.
Vacuum decay	Lifetime route governed by ν_{life} .
Bubble nucleation	Lifetime route for zero-temperature decay or thermal route for finite-temperature nucleation, depending on target.
Thermal phase history	Thermodynamic route governed by ν_{therm} .
Higgs-inflation survival	Richer cosmological-history target or high-scale boundary update.
Planck-scale sensitivity	High-scale/UV boundary condition, singleton-certificate component, or target update.
Absolute vacuum energy	Source-ledger/ZPE obstruction; not a Higgs-vacuum endpoint by itself.
Anthropic or cosmological selection	Richer-scope cosmological source target, not same-scope Higgs endpoint standing.

Definition 7.27 (Quotients as kernels of verdict maps). For each ledger, the endpoint quotient is the kernel of the corresponding verdict map restricted to the same-scope transport orbit:

$$\begin{aligned}
R_1 \sim_{T=0}^{\text{stat}} R_2 &\iff R_1 \mathcal{T}_0^{\text{stat}} R_2 \text{ and } \nu_{\text{stat}}(R_1) = \nu_{\text{stat}}(R_2), \\
L_1 \sim_{T=0}^{\text{life}} L_2 &\iff L_1 \mathcal{T}_0^{\text{life}} L_2 \text{ and } \nu_{\text{life}}(L_1) = \nu_{\text{life}}(L_2), \\
F_1 \sim_{T>0}^{\text{therm}} F_2 &\iff F_1 \mathcal{T}_T^{\text{therm}} F_2 \text{ and } \nu_{\text{therm}}(F_1) = \nu_{\text{therm}}(F_2).
\end{aligned}$$

Thus the quotient is not verdict-circular. The verdict maps are typed status maps, and the quotient relation is their same-scope kernel.

Definition 7.28 (Zero-temperature static quotient). Let $R_i = V_{\text{eff}}^{(g_i, s_i, \mu_i, \mathcal{M}_i)}(\phi_i)$ be two zero-temperature effective-potential representatives inside $D_{\text{HV},0}^{T=0}$. Define

$$R_1 \sim_{T=0}^{\text{stat}} R_2$$

iff R_1 and R_2 are connected by transformations in $\mathcal{T}_0^{\text{stat}}$ and preserve the same static diagnostic verdict over the declared anchor and uncertainty envelope: no deeper admissible branch, lower-branch diagnostic availability, or uncontrolled static diagnostic status. The quotient is $[V_{\text{eff}}]_{T=0}^{\text{stat}}$.

Definition 7.29 (Zero-temperature lifetime quotient). Let L_i be two vacuum-decay/lifetime presentations, including bounce-action, decay-rate, and comparison-four-volume data. Define

$$L_1 \sim_{T=0}^{\text{life}} L_2$$

iff L_1 and L_2 are related by transformations in $\mathcal{T}_0^{\text{life}}$ and preserve the same lifetime verdict relative to the declared criterion: effectively stable/no lower decay channel, metastable with lifetime exceeding the comparison scale, unstable on the comparison scale, or uncontrolled lifetime audit residue. The quotient is $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$; only the physical lifetime verdicts are selectable endpoint classes.

Definition 7.30 (Finite-temperature thermodynamic quotient). Let F_i be two finite-temperature free-energy or dimensionally reduced/lattice/perturbative presentations. Define

$$F_1 \sim_{T>0}^{\text{therm}} F_2$$

iff F_1 and F_2 are connected by transformations in $\mathcal{T}_T^{\text{therm}}$ and preserve the same thermodynamic transition verdict: crossover, first-order transition, second-order/critical point, or uncontrolled same-scope thermal status. The quotient is $[\mathcal{F}_T]_{T>0}^{\text{therm}}$. A new barrier-generating carrier, threshold, scalar sector, or boundary rule exits this quotient and is recorded as a map to $D_{\text{HV},+}$.

Theorem 7.31 (Quotient-construction necessity). *A Higgs-vacuum endpoint claim has standing only as a class in $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, or $[\mathcal{F}_T]_{T>0}^{\text{therm}}$, or as a richer-scope update $D_{\text{HV},+}$.*

Proof. By endpoint-use governance, an endpoint claim must be invariant under admissible redescription of the same target. The three equivalence relations are precisely the admissible redescription relations for the static, lifetime, and thermal ledgers. Any claim outside these quotients either changes under a same-target transformation, in which case it is representative skin, or adds new carrier/boundary content, in which case it is richer scope. Thus endpoint standing requires descent to one of the typed quotient objects. $\square$

8 Gauge-vacuum compatibility and Nielsen-identity mechanics

8.1 Gauge variation and field-coordinate non-standing

The effective potential in gauge theories is not generally a gauge-invariant scalar function of a raw field coordinate. The Nielsen/Fukuda-Kugo identity has the schematic form

$$\left(\xi \frac{\partial}{\partial \xi} + C(\phi, \xi) \frac{\partial}{\partial \phi} \right) V(\phi, \xi) = 0,$$

which means that changing gauge is compensated by a field-coordinate displacement in the representative potential. Fukuda and Kugo showed how gauge dependences of the effective action and potential are controlled, and concluded that values at solutions of the effective equations can be gauge invariant when handled properly [8]. The Standard Model instability field value, however, is not such an endpoint object. Di Luzio and Mihaila emphasize that the field value at which V_{eff} turns negative is gauge dependent even though the stability bound can be formulated in a gauge-independent way [9].

Definition 8.1 (Gauge-vacuum compatible endpoint). A Higgs-vacuum endpoint class C is gauge-vacuum compatible when its endpoint content is invariant under admissible gauge redescription or is expressed through a gauge-controlled physical predicate such as a properly defined stability bound, decay/lifetime verdict, or thermodynamic transition class.

Theorem 8.2 (Field-coordinate instability non-standing). *A field coordinate ϕ_* , or a displayed “instability scale” inferred from a field value where a representative potential turns negative, has no primitive endpoint standing unless it descends to a gauge-independent or gauge-controlled endpoint predicate.*

Proof. The Nielsen identity shows that gauge changes move the representative potential along field-coordinate directions. Therefore the field coordinate of a crossing or extremum is not a gauge-invariant endpoint datum. If a calculation maps the representative into a gauge-controlled stability bound, decay rate, or lifetime class, then the mapped predicate can carry standing. But the raw ϕ_* cannot. It is a coordinate in a representative chart, not the endpoint. $\square$

Theorem 8.3 (Gauge-vacuum compatibility requirement). *A Higgs-vacuum endpoint class has standing only if it is gauge-vacuum compatible. Raw gauge-fixed extrema, field-coordinate instability locations, and representative-specific potential features are diagnostics, not primitive endpoints.*

Proof. If a claim changes under an admissible gauge redescription while no physical witness changes, then the claim is representative skin. If the transformation changes a physical witness, then it was not a same-scope gauge redescription and must be handled as a target update or witness change. Hence only gauge-vacuum compatible predicates can carry endpoint standing. $\square$

9 Renormalization, scheme, and matching transport

9.1 RG as a standing-preserving transport ledger

Running couplings are essential for high-scale Higgs-vacuum analysis. The Higgs quartic $\lambda(\mu)$, top Yukawa $y_t(\mu)$, electroweak couplings, and $\alpha_s(\mu)$ are the machinery by which measured low-energy anchors are transported to high field and high scale. But this machinery is not the endpoint.

Definition 9.1 (Higgs RG/matching transport ledger). $\text{RGTrans}_{\text{HV}}$ is the ledger of admissible maps relating measured anchors, Lagrangian parameters, running couplings, threshold matchings, scheme conventions, EFT envelopes, and high-scale extrapolations inside $D_{\text{HV},0}$.

Theorem 9.2 (RG diagnostic non-promotion). *A running-coupling diagnostic such as $\lambda(\mu) < 0$, a zero-crossing scale, or a near-critical RG trajectory is not by itself a Higgs-vacuum endpoint. It becomes endpoint-relevant only through a standing-preserving map to a gauge-, scheme-, matching-, and lifetime-compatible classification.*

Proof. The value of a running parameter is a coordinate in a scale and scheme representation. A scheme change, threshold treatment, or matching convention can alter displayed running values while preserving physical predictions. Therefore $\lambda(\mu)$ and its zero crossing cannot be identical to the endpoint. They are diagnostics. Once transported through a valid stability/lifetime calculation with declared uncertainties, they can support an endpoint claim; without that transport, they remain representative data. $\square$

9.2 Top-mass anchor non-identity

The top mass is the most dangerous anchor because it strongly affects the RG evolution of the Higgs quartic. The most precise collider combinations are built from direct reconstruction methods whose mass parameter is closely related to, but not identical by definition with, the pole or short-distance mass required in high-order vacuum-stability calculations. Hoang’s review stresses that direct measurements quote a Monte-Carlo generator mass parameter m_t^{MC} and that a rigorous relation to field-theoretic mass schemes involves unresolved perturbative and nonperturbative issues [3].

Theorem 9.3 (Top-mass anchor non-identity). *A measured direct-reconstruction top-mass anchor is not automatically identical to the short-distance or pole-mass input used in high-scale Higgs-vacuum RG transport. A zero-temperature endpoint that depends on m_t cannot be selected without a top-mass conversion and matching certificate.*

Proof. The measured top anchor enters the endpoint analysis through y_t and mass-scheme matching. If the measured mass parameter and the field-theory mass input are treated as identical without a conversion certificate, then an unlicensed identification has been inserted into the endpoint proof. If the conversion is supplied, it is part of the singleton certificate. If it is not supplied, the endpoint selection remains load-bearing on an unclosed anchor-transport component. $\square$

Top-mass route	What it supplies	AASC status
Direct reconstruction / Monte-Carlo mass	High-precision collider anchor tied to event-generator modelling	Anchor datum requiring conversion before use as RG mass input.
Pole-mass extraction	Field-theory mass parameter with perturbative and renormalon-sensitive interpretation	Usable when extraction and uncertainty model are declared.
$\overline{\text{MS}}$ or short-distance mass	Scheme-defined running mass suited to RG transport	Requires matching from measured anchor and scheme declaration.
Inclusive cross-section extraction	Less direct but more field-theory-connected mass inference	Alternative anchor route with larger/structured uncertainties.
Future threshold scan	Potentially cleaner short-distance mass route in a lepton collider threshold environment	Future source-state update, not retroactive same-target closure.

10 Effective-potential diagnostics and lower-branch burden

10.1 Positive admission of effective potentials

Theorem 10.1 (Effective-potential diagnostic admission). *The Higgs effective potential is an admissible diagnostic representative of the Higgs-vacuum endpoint target when used through a gauge-, scheme-, RG-, and matching-controlled transport map. It is not primitive endpoint standing in its raw representative form.*

Proof. Effective potentials encode the field-dependent vacuum response of the theory and are indispensable for practical stability, tunneling, and thermal analyses. Their admissible role is diagnostic because their representative form depends on gauge, scheme, scale, and matching choices. The endpoint role is obtained only after the representative is transported into $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, or $[\mathcal{F}_T]_{T>0}^{\text{therm}}$. Thus the theorem admits effective-potential use while blocking representative promotion. $\square$

Theorem 10.2 (Effective-potential quotient theorem). *A Higgs-vacuum statement derived from $V_{\text{eff}}^{(g,s,\mu,\mathcal{M})}$ has endpoint standing only if the statement descends to $[V_{\text{eff}}]_{T=0}^{\text{stat}}$ or $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, or to $[\mathcal{F}_T]_{T>0}^{\text{therm}}$ in the finite-temperature case.*

Proof. Suppose the statement depends on a representative feature that changes under an admissible gauge, RG, scheme, matching, or field-redescription transformation. Then it fails same-target invariance and cannot be endpoint standing. If it survives those transformations or is mapped into a physical lifetime or thermodynamic predicate that survives them, it belongs to the relevant quotient. Hence quotient descent is necessary. $\square$

10.2 Lower branch

Lemma 10.3 (Endpoint-function exhaustion for lower Higgs branches). *By Theorem 7.23, let B be a lower branch of a Higgs effective-potential representative admitted as a same-scope diagnostic*

in $D_{\text{HV},0}^{T=0}[\mathbf{S}_{2026}]$. Any claim that B is physically endpoint-bearing must change at least one of the following ledgers:

1. the static ordering ledger, by deciding whether a deeper admissible same-scope branch survives;
2. the lifetime ledger, by deciding whether an admitted lower channel yields metastability or instability relative to $\mathcal{L}$;
3. the finite-temperature thermodynamic ledger, by supplying a transition-order verdict in $D_{\text{HV},0}^{T>0}$;
4. the carrier/source inventory, by adding a threshold, field, symmetry, gravitational boundary, or UV rule and thereby moving to $D_{\text{HV},+}$.

If a purported lower-branch claim changes none of these ledgers, it is status-inert: it has no gauge-stable static verdict, no decay or lifetime verdict, no thermal transition verdict, and no new carrier/source certificate. Such a claim is diagnostic availability only, not endpoint standing.

Proof. The four ledgers are exactly the verdict-map completions established in Theorem 7.23, and Lemma 7.24 excludes any fifth endpoint function. Static ordering is the same-scope zero-temperature way for a branch to affect whether the electroweak vacuum has a deeper competitor. Lifetime status is the same-scope zero-temperature way for a surviving lower channel to decide false-vacuum fate. Finite-temperature transition order is a different target ledger, already separated as $D_{\text{HV},0}^{T>0}$. Any new field, threshold, symmetry, gravitational assumption, or UV boundary changes the carrier/source inventory and is therefore richer scope. A claim that affects none of these functions has no endpoint-relevant consequence left; treating it as endpoint standing would promote a representative branch without a standing rule. $\square$

Lemma 10.4 (Lower-branch route normal form). *Let C_{lower} be the candidate claim that a lower branch of a Higgs effective-potential representative carries primitive same-scope endpoint standing. Any claim-bearing same-scope use of that branch must answer one of four status questions:*

1. *Does the branch fail to survive static and lifetime audit, yielding absolute-stability status C_{abs} ?*
2. *Does the branch survive as a decay channel whose decay probability stays below the declared comparison threshold, yielding metastability C_{meta} ?*
3. *Does the branch survive as a decay channel whose decay probability exceeds the declared comparison threshold, yielding instability C_{unst} ?*
4. *Does its standing require new carrier, boundary, threshold, gravitational, or UV structure, yielding richer-scope update $D_{\text{HV},+}$?*

No additional primitive lower-vacuum role remains inside $D_{\text{HV},0}^{T=0}[\mathbf{S}_{2026}]$.

Proof. The endpoint-function exhaustion lemma fixes the possible functions of a lower branch. In the zero-temperature same-scope target, the static quotient $[V_{\text{eff}}]_{T=0}^{\text{stat}}$ decides whether the branch survives as an admissible lower-channel diagnostic or collapses to no deeper same-scope branch or uncontrolled static audit residue. If no lower channel survives the static and lifetime audit, the physical endpoint route is absolute stability C_{abs} . If a lower channel survives, endpoint use must pass through the lifetime quotient $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, whose physical verdicts are metastability and instability according to the declared decay-probability threshold over $\mathcal{V}_4$. If the branch requires a new scalar, threshold, symmetry, gravitational assumption, UV boundary, or other carrier/source to become standing-bearing, the target is updated to $D_{\text{HV},+}$. A thermal transition-order role is not a fifth zero-temperature route; it belongs to $D_{\text{HV},0}^{T>0}$, where the measured minimal Standard Model selects crossover. Lemma 7.24 rules out remaining alleged endpoint functions. Thus the only

same-target alternatives are the first three endpoint classes, and every carrier/source repair is richer scope. $\square$

Theorem 10.5 (Primitive lower-branch claim-form deletion without definitional exclusion). *Let C_{lower} denote the claim-form that a representative lower branch itself has primitive same-scope endpoint standing in $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$, rather than standing obtained through $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, C_{abs} , C_{meta} , C_{unst} , or $D_{\text{HV},+}$. Then C_{lower} has no certified same-scope survivor:*

$$\text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0.$$

A lower branch of a representative effective potential is diagnostic availability. It becomes endpoint-bearing only by descending to C_{abs} , C_{meta} , C_{unst} , or $D_{\text{HV},+}$ through the appropriate quotient and certificate.

Proof. The theorem statement defines C_{lower} as a claim-form rather than as a physical calculation. The lower branch is presented in a gauge, scheme, RG, matching, perturbative, and field-coordinate chart, so it is not identical to an endpoint verdict. The lower-branch route-normal-form lemma exhausts every standing-bearing use of the branch: absolute-stability status, metastability/lifetime status, instability/lifetime status, or richer-scope update. In the first three cases the selected object is C_{abs} , C_{meta} , or C_{unst} , not C_{lower} . In the fourth case the target is $D_{\text{HV},+}$, not the same-scope target. No fifth primitive role survives the no-promotion criterion by Lemma 7.24. Therefore C_{lower} is deleted by route exhaustion rather than by definition. $\square$

Theorem 10.6 (Non-tautological lower-branch burden). *Every standing-bearing promotion of a representative lower branch must enter exactly one of four endpoint-status routes:*

$$C_{\text{abs}}, \quad C_{\text{meta}}, \quad C_{\text{unst}}, \quad D_{\text{HV},+}.$$

There is no fifth primitive lower-vacuum endpoint route inside $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$.

Proof. This is the survivor version of the route-normal-form lemma. Static audit either deletes the lower branch as a physical lower channel, admits it only as a diagnostic residue, or leaves it uncontrolled. Lifetime audit turns an admitted lower decay channel into metastability or instability according to the declared threshold. New carrier/source structure changes target. A branch outside those cases supplies no gauge-invariant field coordinate, no lifetime verdict, no stability verdict, and no new carrier certificate. It is therefore only diagnostic availability promoted without a standing rule. By no-promotion and endpoint selection, that route is deleted. $\square$

Theorem 10.7 (Physics-idiom translation). *The AASC statement $\text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0$ does not deny the standard conditional idiom “the electroweak vacuum is metastable under specified inputs.” It translates that idiom as a certificate-dependent claim of the form*

$$\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}}) \Rightarrow \text{Sel}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}}).$$

It denies only the stronger unlicensed promotion

$$\text{LowerBranch}(V_{\text{eff}}^{(g,s,\mu,\mathcal{M})}) \Rightarrow \text{PrimitiveStanding}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{lower}}).$$

Proof. The standard metastability claim is a lifetime-class claim. It becomes endpoint-bearing when the mass inputs, gauge treatment, RG/scheme matching, perturbative order, high-scale assumptions, uncertainty model, and lifetime threshold are declared. That is precisely the singleton-certificate route. The deleted object is different: a raw lower branch treated as the endpoint before the certificate is supplied. The theorem therefore preserves conditional metastability analyses while deleting primitive lower-branch standing. $\square$

Corollary 10.8 (Lower-branch non-promotion).

$$\text{LowerBranch}(V_{\text{eff}}^{(g,s,\mu,\mathcal{M})}) \not\Rightarrow \text{PrimitiveStanding}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C).$$

Corollary 10.9 (False-vacuum normal form). *A standing-bearing false-vacuum claim in $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$ has the normal form*

$$([V_{\text{eff}}]_{T=0}^{\text{stat}}, [\Gamma_{\text{decay}}]_{T=0}^{\text{life}}, A_{\text{HV}}, \text{RGTrans}_{\text{HV}}, \mathcal{L}),$$

where $\mathcal{L}$ is the declared lifetime or comparison-four-volume criterion.

Lemma 10.10 (No one-step retyping from lower branch to vacuum fate). *The proof never uses a shortcut of the form*

$$\text{LowerBranch}(V_{\text{eff}}) \Rightarrow \text{Sel}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}})$$

or

$$\lambda(\mu) < 0 \Rightarrow \text{Sel}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}}).$$

The typed route is

$$V_{\text{eff}}^{(g,s,\mu,\mathcal{M})} \rightarrow [V_{\text{eff}}]_{T=0}^{\text{stat}} \rightarrow [\Gamma_{\text{decay}}]_{T=0}^{\text{life}} \rightarrow \mathcal{S}_{\text{HV}}^{T=0}(C) \rightarrow \text{Sel}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C),$$

with gauge, RG/scheme, matching, top-mass, perturbative, high-scale, and lifetime controls checked before selection.

Proof. A lower branch and a running-coupling crossing are diagnostic inputs. They do not by themselves specify a gauge-stable field-space object, a mass-scheme-stable top anchor, a lifetime criterion, or a rival-deletion rule. The intermediate quotients and singleton certificate supply exactly those missing status transitions. Omitting them confuses diagnostic availability with endpoint selection. $\square$

11 Zero-temperature singleton burden

11.1 Anchor-closed singleton certificate

Definition 11.1 (Anchor-closed zero-temperature singleton certificate). A zero-temperature singleton certificate for endpoint class $C \in \{C_{\text{abs}}, C_{\text{meta}}, C_{\text{unst}}\}$ is

$$\mathcal{S}_{\text{HV}}^{T=0}(C) = (A_{\text{HV}}, \mathcal{G}, \mathcal{R}, \mathcal{M}_t, \mathcal{P}, \mathcal{U}, \mathcal{L}, \Delta),$$

where:

- A_{HV} is the measured Higgs/top/ α_s /electroweak anchor ledger;
- $\mathcal{G}$ is the gauge-control certificate;
- $\mathcal{R}$ is the RG/scheme transport certificate;

- $\mathcal{M}_t$ is the top-mass conversion and matching certificate;
- $\mathcal{P}$ is the perturbative-order and truncation-uncertainty certificate;
- $\mathcal{U}$ is the declared high-scale, Planck-scale, EFT, and gravity-assumption boundary;
- $\mathcal{L} = (\mathcal{V}_4, \epsilon, \Pi_{\Gamma_4}, \mathcal{E}_{\text{life}}, \mathcal{U}_{\text{grav}})$ is the lifetime or stability comparison certificate;
- Δ is the declared confidence or separation threshold that eliminates inequivalent rival endpoints.

Theorem 11.2 (Stable/metastable singleton burden). *A zero-temperature endpoint selection*

$$\text{Sel}_{D_{\text{HV},0}^{T=0}}(C_{\text{abs}}) \quad \text{or} \quad \text{Sel}_{D_{\text{HV},0}^{T=0}}(C_{\text{meta}})$$

requires an anchor-closed singleton certificate $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{abs}})$ or $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}})$.

Proof. By the endpoint-selection criterion, selecting C_{abs} or C_{meta} requires certified standing for the selected class and certified deletion of inequivalent rivals. Gauge control is needed to prevent a gauge-representative branch from selecting the endpoint. RG/scheme transport is needed because high-scale stability depends on running quantities. Top-mass conversion is needed because the top anchor enters through a mass-scheme-dependent Yukawa input. Perturbative-order and uncertainty control are needed to determine whether the endpoint remains stable under truncation and input variation. High-scale assumptions are needed because extrapolation to very high field values can be altered by thresholds, EFT breakdown, or gravity-sensitive corrections. A lifetime rule is needed for metastability. A confidence or separation threshold is needed to delete rival classes. These are exactly the components of $\mathcal{S}_{\text{HV}}^{T=0}(C)$. $\square$

Lemma 11.3 (Stable/metastable rival-basin non-deletion). *If the source-state anchor and uncertainty envelope intersects both an absolute-stability basin and a metastability basin under the declared high-scale transport map, then neither C_{abs} nor C_{meta} is selected unless $\mathcal{S}_{\text{HV}}^{T=0}(C)$ supplies the separation threshold Δ deleting the rival basin.*

Proof. This is the zero-temperature specialization of rival-basin non-deletion. The stability and metastability basins are inequivalent endpoint classes. If the admitted anchor/matching/uncertainty envelope reaches both, the endpoint-selection criterion cannot certify unique survivor status for either. Deleting the rival requires the missing component of $\mathcal{S}_{\text{HV}}^{T=0}(C)$: a declared mass conversion, RG/matching, perturbative, high-scale, and statistical separation certificate. $\square$

Theorem 11.4 (Source-state certificate failure). *The source-state anchor ledger $\mathcal{S}_{2026}$ does not by itself contain $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{abs}})$ or $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}})$ as a purely structural AASC certificate. Therefore neither absolute stability nor metastability is selected as a fixed-scope zero-temperature structural singleton in $D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]$.*

Proof. Source-state high-order analyses still identify the top mass and strong coupling as central load-bearing inputs for Standard Model vacuum stability, and estimate that reducing their uncertainties by a factor of two to three would be sufficient to establish or refute Standard Model stability at the 5σ level [6]. The top-mass interpretation itself requires a conversion from direct-reconstruction or Monte-Carlo mass parameters to field-theoretic masses used in RG calculations [3]. The gauge-dependence literature separates physical stability bounds from gauge-dependent instability field values [9]. These facts show that the source-state anchor ledger lacks at least one load-bearing closure component of $\mathcal{S}_{\text{HV}}^{T=0}(C)$ for a purely structural endpoint selection. Hence $\text{Audit}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(D_{\text{HV},0}) = \text{PrecisionSensitive}$ rather than $\text{Sel}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(C_{\text{abs}}) = 1$ or $\text{Sel}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(C_{\text{meta}}) = 1$. $\square$

Source-state update. A later source state that closes the top-mass, strong-coupling, matching, perturbative-order, and high-scale boundary package may select C_{abs} , C_{meta} , or C_{unst} through ν_{stat} or ν_{life} . Such a result would instantiate a new source-state theorem rather than refuting the theorem proved for $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$. The present result deletes only primitive lower-branch endpoint standing and blocks singleton selection by the S_{2026} ledger alone.

Corollary 11.5 (Numerical results are preserved). *Numerical vacuum-stability analyses can become endpoint-bearing when their inputs, mass schemes, matching, perturbative order, gauge treatment, lifetime criterion, and uncertainty model are declared. The theorem blocks only retyping those conditional numerical results as structural singleton selection without the certificate.*

12 Finite-temperature endpoint: the positive same-scope result

12.1 Thermal target

The finite-temperature electroweak question belongs to $D_{\text{HV},0}^{T>0}$ and is represented by $[\mathcal{F}_T]_{T>0}^{\text{therm}}$. Its same-scope endpoint classes are thermodynamic: crossover, first-order transition, or second-order/critical transition. Richer-scope behavior is not a fourth same-scope verdict; it is a target update to $D_{\text{HV},+}^{T=0}$. This target is distinct from $D_{\text{HV},0}^{T=0}$.

Theorem 12.1 (Imported physical-Higgs Standard Model crossover). *With the physical Higgs mass and the measured minimal Standard Model field content, the electroweak finite-temperature transition is a smooth crossover rather than a first-order transition.*

Proof. This is the standard finite-temperature input. D’Onofrio and Rummukainen’s lattice study reports a smooth crossover with $T_c = 159.5 \pm 1.5$ GeV and a width of about 5 GeV [13]. Laine and Meyer describe Standard Model thermodynamics across the electroweak crossover and state that the physical-Higgs Standard Model has no bulk phase transition but a soft point near 160 GeV [14]. AASC does not derive the lattice result; it imports it as the physical comparison theorem. $\square$

Corollary 12.2 (Fixed-scope electroweak crossover endpoint). *In $D_{\text{HV},0}^{T>0}$, the selected finite-temperature endpoint is crossover:*

$$\text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1.$$

Proof. $D_{\text{HV},0}^{T>0}$ contains the measured minimal Standard Model, the thermodynamic verdict map ν_{therm} , and no additional thermal barrier-generating carrier. The imported physical theorem supplies the same-scope thermodynamic verdict. Therefore the endpoint quotient $[\mathcal{F}_T]_{T>0}^{\text{therm}}$ selects C_{cross} . $\square$

Corollary 12.3 (Same-scope first-order deletion). *Within $D_{\text{HV},0}^{T>0}$,*

$$\text{Surv}_{D_{\text{HV},0}^{T>0}}(C_{1\text{st}}) = 0.$$

Proof. The imported physical-Higgs Standard Model theorem classifies the measured minimal Standard Model thermal endpoint as crossover. First-order transition is an inequivalent value of the thermodynamic verdict map ν_{therm} . Since $D_{\text{HV},0}^{T>0}$ contains no additional carrier or source structure beyond the measured minimal Standard Model, no same-scope representative in $[\mathcal{F}_T]_{T>0}^{\text{therm}}$ supplies the first-order verdict. Barrier and latent-heat language explains the physical distinction, but the load-bearing inference is the imported crossover theorem plus quotient inequivalence. Any first-order verdict therefore requires an updated thermodynamic target with new carrier/source structure, and $C_{1\text{st}}$ has no same-scope survivor in $D_{\text{HV},0}^{T>0}$. $\square$

13 First-order EWPT richer-scope burden

Definition 13.1 (Richer-scope Higgs-vacuum certificate). A richer-scope Higgs-vacuum update has the normal form

$$\mathcal{C}_{\text{HV},+} = (\Sigma_{\text{new}}, D_{\text{HV},0} \rightarrow D_{\text{HV},+}, \text{Transport}_+, \text{ThermalPotential}_+, \\ \text{Stability}_+, \text{AnchorDescent}_+, \text{Witness}_+).$$

The new carrier Σ_{new} may be a scalar singlet, second Higgs doublet, scalar triplet, supersymmetric sector, portal field, higher-dimensional operator residue, nonminimal gravitational coupling, UV threshold, or other specified structure.

Theorem 13.2 (First-order electroweak phase-transition richer-scope burden). *Any first-order electroweak phase-transition claim incompatible with the measured minimal-SM crossover class requires a richer-scope certificate $\mathcal{C}_{\text{HV},+}$.*

Proof. The same-scope measured minimal Standard Model selects crossover. A first-order claim therefore must alter at least one thermodynamic ingredient: the carrier inventory, scalar potential, thermal cubic structure, threshold matching, EFT operator content, or UV/gravity boundary. If no such alteration is specified, the claim has no source of barrier formation inside $D_{\text{HV},0}^{T>0}$. If it is specified, the target is no longer $D_{\text{HV},0}^{T>0}$ but $D_{\text{HV},+}$. Thus the richer-scope certificate is necessary. $\square$

Table 4: Representative richer-scope Higgs-vacuum routes and the new load required for endpoint re-entry.

Route	New load	Endpoint role
Singlet scalar	Extra scalar and portal coupling	Can stabilize high-field running or generate a thermal barrier.
Two-Higgs-doublet model	Additional Higgs doublet and scalar potential	Can modify CP, vacuum structure, and thermal transition order.
Scalar triplet	New electroweak representation	Can change thermal cubic terms and phase-transition strength.
Supersymmetric thresholds	Scalar partners and threshold corrections	Modify matching and high-scale running.
Higgs portal dark sector	Hidden scalar/vector load coupled to $H^\dagger H$	Can stabilize or alter thermal history.
SMEFT barrier	Higher-dimensional operator residue	Can create a barrier or shift the thermal endpoint.
Gravity/Higgs-curvature	Nonminimal metric or Planck-scale load	Alters high-field extrapolation boundary.
Asymptotic-safety route	UV fixed-point structure	Changes high-scale RG assumptions.

14 Same-scope measured minimal-SM electroweak-baryogenesis route deletion

Definition 14.1 (Measured minimal Standard Model EWBG target). $D_{\text{EWBG},0}$ is the same-scope electroweak-baryogenesis route target induced by $D_{\text{HV},0}^{T>0}$ together with the minimal Standard Model

CP, sphaleron, transport, and observed-asymmetry ledgers. It contains:

1. the measured minimal Standard Model finite-temperature Higgs/electroweak target;
2. electroweak baryon-number violation through the Standard Model sphaleron sector;
3. Standard Model CKM CP as the only CP source;
4. a success predicate requiring production of the observed baryon asymmetry and preservation against washout within the declared electroweak route;
5. the usual washout/out-of-equilibrium route conditions for electroweak baryogenesis;
6. no extra scalar, portal, CP-violating sector, wall-source carrier, threshold, or UV extension.

$D_{\text{EWBG},+}$ is the corresponding richer-scope update target.

Definition 14.2 (Observed-asymmetry success predicate). The symbol η_B^{obs} denotes the admitted observed-baryon-asymmetry success predicate for the electroweak-baryogenesis route target. It is not a new numerical calculation in this paper. It is the target condition for route success, not an assumption that the route succeeds. It records that a successful route must produce an admissible baryon asymmetry with the observed sign and magnitude class and must preserve it against washout under the declared thermal and transport ledger. A same-scope route that lacks a load-bearing production or preservation component fails this predicate.

Definition 14.3 (Same-scope minimal-SM observed-asymmetry EWBG route). The route EWBG_{SM} is the claim that the measured minimal Standard Model itself supplies a successful electroweak-baryogenesis route producing and preserving the observed baryon asymmetry within $D_{\text{EWBG},0}$. Its route tuple is stated without building the failure into the definition:

$$\text{EWBG}_{\text{SM}} = (\mathcal{B}_{\text{sph}}, \text{CPsource}^{\text{SM}}, \text{Departure}_{\text{EW}}, \text{Transport}, \text{WashoutAvoidance}, \eta_B^{\text{obs}}).$$

Here η_B^{obs} denotes the observed-asymmetry success ledger, and $\text{Departure}_{\text{EW}}$ denotes the electroweak departure-from-equilibrium component required by the route, not a stipulated first-order verdict. A missing load-bearing component deletes the route as a same-scope survivor.

The electroweak-baryogenesis input used here is standard. Reviews describe EWBG as requiring electroweak baryon-number violation, CP violation, and an out-of-equilibrium electroweak transition with washout avoidance, and they emphasize that Standard Model extensions are normally invoked to obtain a sufficiently strong transition and new CP/source structure [17, 18]. A baryon-asymmetry computation is theorem-relevant here only if it supplies, breaks, or updates an admitted route condition or the source-state ledger; the AASC result classifies the same-scope minimal-Standard-Model route after those route conditions are imported.

Theorem 14.4 (Imported minimal-SM electroweak-baryogenesis failure). *In the measured minimal Standard Model, the electroweak-baryogenesis route does not supply a successful production-and-preservation route for the observed baryon asymmetry. The imported failure has two independent limbs: the physical-Higgs electroweak thermal history is crossover rather than a strong first-order transition, and the CKM phase alone is not an adequate electroweak-baryogenesis CP/source.*

Proof. The thermal limb is the physical-Higgs crossover theorem already imported in the finite-temperature section. The CP/source limb is the standard measured minimal-SM EWBG insufficiency result: CKM-only mechanisms fail once the relevant damping/decoherence and source-size limits are accounted for, and the Standard Model falls short without enlarging the symmetry-breaking sector and adding new CP/source structure [19, 20]. Broad EWBG reviews organize the rescue class in exactly this way, through new scalar/transition structure and new CP/source transport [17, 18]. $\square$

Lemma 14.5 (Measured-SM electroweak departure normal form). *In the measured minimal-Standard-Model electroweak route, the electroweak departure component that can support standard EWBG is the strong-first-order, bubble-wall route or an equivalent out-of-equilibrium electroweak source. The physical-Higgs minimal Standard Model supplies crossover instead. Therefore $\text{Departure}_{\text{EW}}$ is not certified in $D_{\text{EWBG},0}$.*

Proof. Successful electroweak baryogenesis of the observed asymmetry requires baryon-number violation, CP violation, a departure from thermal equilibrium, transport/source conversion, and washout avoidance. In the measured minimal Standard Model target, the relevant electroweak departure channel would have to come from the electroweak thermal history itself. The imported physical theorem and the AASC endpoint corollary give crossover, not a first-order bubble-wall transition, and $D_{\text{EWBG},0}$ contains no additional out-of-equilibrium carrier. Thus the departure component is absent as a same-scope route condition without defining EWBG_{SM} as first-order by fiat. $\square$

Lemma 14.6 (No route laundering from zero-temperature vacuum structure). *A zero-temperature lower-branch diagnostic or metastability claim does not supply the missing electroweak departure, bubble-wall dynamics, or CP/source transport required for electroweak baryogenesis:*

$$C_{\text{lower}} \not\Rightarrow \text{Departure}_{\text{EW}}, \quad C_{\text{meta}} \not\Rightarrow \text{Departure}_{\text{EW}}, \quad C_{\text{meta}} \not\Rightarrow \text{CPsource}_{\text{EWBG}}^{\text{adequate}}.$$

Proof. The zero-temperature target $D_{\text{HV},0}^{T=0}$ classifies static and lifetime structure. The electroweak-baryogenesis route target $D_{\text{EWBG},0}$ requires a finite-temperature out-of-equilibrium route and an adequate CP/source ledger. These are different target coordinates. A lower zero-temperature branch may be relevant to high-scale stability, but it does not create thermal bubble walls, latent heat, sphaleron washout avoidance, or a new CP source. Therefore no zero-temperature false-vacuum diagnostic can launder itself into an EWBG route component. $\square$

Corollary 14.7 (AASC same-scope measured minimal-SM EWBG route deletion). *In the measured minimal Standard Model EWBG target,*

$$\text{Surv}_{D_{\text{EWBG},0}}(\text{EWBG}_{\text{SM}}) = 0.$$

Proof. The imported measured minimal-SM EWBG failure theorem supplies the physical route facts: the electroweak departure limb fails because the finite-temperature endpoint is crossover, and the adequate CP/source limb fails because CKM-only CP does not supply a successful observed-asymmetry source. The definition of $D_{\text{EWBG},0}$ contains no additional out-of-equilibrium carrier, scalar sector, CP/source ledger, or richer transport channel. Therefore the route tuple EWBG_{SM} lacks load-bearing components needed before η_B^{obs} can be met. By the endpoint-selection criterion, the successful observed-asymmetry route has no certified same-scope survivor. $\square$

Definition 14.8 (Richer-scope EWBG certificate). A successful electroweak-baryogenesis update has the normal form

$$\begin{aligned} C_{\text{EWBG},+} = & (\Sigma_{\text{new}}, \text{Departure}_{\text{EW},+}, \text{CPsource}_{+}, \text{Sphaleron}_{+}, \\ & \text{Transport}_{+}, \text{AnchorDescent}_{+}, \text{Witness}_{+}). \end{aligned}$$

The certificate must specify the new carrier or law, the out-of-equilibrium electroweak route, the CP/source mechanism, the sphaleron/washout condition, transport through the bubble wall or equivalent dynamics, descent to measured Higgs/top/electroweak anchors, and possible witnesses such as collider signatures, EDMs, gravitational waves, flavor signals, or cosmological residues.

Corollary 14.9 (EWBG richer-scope recognition). *Any successful electroweak-baryogenesis route compatible with the measured Higgs sector but not contained in $D_{\text{EWBG},0}$ is a richer-scope update $D_{\text{EWBG},+}$ requiring $C_{\text{EWBG},+}$.*

Proof. The same-scope route is deleted. A successful route must therefore add at least one missing load-bearing element: a first-order transition/barrier or alternative out-of-equilibrium source, new CP/source structure, modified transport, or a changed sphaleron/washout ledger. Each such addition changes the carrier or law inventory and is therefore $D_{\text{EWBG},+}$, not a same-scope survivor of $D_{\text{EWBG},0}$. $\square$

15 Zero-point-energy quotient obstruction

15.1 Main-text firewall

Higgs-vacuum discussion often tempts a second illegitimate promotion:

$$V_{\text{eff}}(\phi_{\min}) \Rightarrow \Lambda_{\text{cosmo}}.$$

This paper explicitly blocks that move. Renormalized QFT permits additive vacuum-energy counterterms and finite shifts. The absolute homogeneous offset of an effective potential is not a standalone cosmological prediction. What can survive are relational quantities: tunneling actions, phase/free-energy differences, latent heat in a first-order transition, and lifetime classifications.

Theorem 15.1 (Absolute vacuum-energy offset non-promotion). *The absolute offset of the Higgs effective potential is not a standing-bearing cosmological-constant prediction. Higgs-vacuum stability permits relational endpoint claims, tunneling/lifetime claims, and finite-temperature free-energy comparisons, but not direct promotion of a renormalized potential offset into an absolute cosmological vacuum-energy value.*

Proof. A common additive shift of the potential leaves nongravitational transition rates, relative depths, and properly defined thermal comparisons unchanged. The absolute homogeneous offset is altered by renormalization convention and gravitational counterterm choice. A quantity that changes under admissible finite renormalization while no relational witness changes cannot be a standalone prediction. It is boundary input or source-ledger data, not endpoint standing. $\square$

16 Endpoint theorem set

Theorem 16.1 (Endpoint-use governance). *Every claim-bearing Higgs-vacuum endpoint use instantiates the kernel roles of reference, standing, admissibility, and irreversibility.*

Theorem 16.2 (Typed quotient endpoint). *Higgs-vacuum endpoint objects are classes in $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, or $[\mathcal{F}_T]_{T>0}^{\text{therm}}$, or richer-scope updates $D_{\text{HV},+}$.*

Theorem 16.3 (Same-domain route-availability exhaustion). *Candidate metastability protocols in $D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]$ are first classified by route status:*

$$\text{CandRoute}_{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) \Rightarrow \rho_{\text{route}}^{D_{\text{HV},0}^{T=0}[\mathcal{S}_{2026}]}(\mathcal{P}) \in \left\{ \begin{array}{l} \text{DiagnosticOnly, RouteIncomplete, RouteNormal,} \\ \text{TargetUpdate, StandingSmuggling} \end{array} \right\}.$$

Surviving route availability is reserved for route-normal protocols and factors through the partial lifetime normalizer:

$$\begin{aligned} \text{RouteAvail}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}}; \mathcal{P}) = 1 &\Rightarrow \text{NF}_{\text{life}}(\mathcal{P}) \downarrow \\ \wedge \mathcal{P} &\cong_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]} \text{NF}_{\text{life}}(\mathcal{P}). \end{aligned}$$

The older display $\text{RouteAvail}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(C_{\text{meta}}) = \text{RouteAvail}_{\mathcal{R}_{\text{life}}}(C_{\text{meta}})$ is retained only as shorthand for this typed route-form statement; it is not a standing-closure claim.

Corollary 16.4 (Fifth-function closure entails primitive lower-branch deletion). *Once the verdict maps and target-update route are exhausted, failure to supply a standing fifth endpoint function directly deletes primitive lower-branch endpoint standing:*

$$\neg \exists F_5(B) \wedge \text{Exhaust}(\nu_{\text{stat}}, \nu_{\text{life}}, \nu_{\text{therm}}, D_{\text{HV},+}) \Rightarrow \text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0.$$

Theorem 16.5 (Primitive lower-vacuum deletion). *By verdict-map completeness, no-fifth-function closure, and Corollary 7.26, the candidate primitive lower-vacuum class has no same-scope survivor:*

$$\text{Surv}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}^{\text{prim}}(C_{\text{lower}}) = 0.$$

A displayed lower effective-potential branch has diagnostic availability, $\text{DiagnosticAvailability} = 1$. It does not have primitive endpoint standing, $\text{PrimitiveStanding} = 0$, unless it descends through a quotient-stable static, lifetime, thermal, or richer-scope certificate.

Theorem 16.6 (Stable/metastable singleton burden). *Selecting C_{abs} or C_{meta} at zero temperature requires $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{abs}})$ or $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}})$.*

Theorem 16.7 (Source-state zero-temperature audit). *The source-state measured-anchor ledger yields*

$$\text{Audit}_{D_{\text{HV},0}^{T=0}[\text{S}_{2026}]}(D_{\text{HV},0}) = \text{PrecisionSensitive},$$

not structural singleton selection of C_{abs} or C_{meta} .

Theorem 16.8 (Minimal-SM finite-temperature endpoint).

$$\text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1, \quad \text{Surv}_{D_{\text{HV},0}^{T>0}}(C_{\text{1st}}) = 0.$$

Same-scope measured minimal-SM EWBG route deletion. The endpoint theorem package records the earlier imported measured minimal-SM electroweak-baryogenesis failure theorem and the AASC same-scope route-deletion corollary; it does not introduce a second independent proof. The recorded endpoint is

$$\text{Surv}_{D_{\text{EWBG},0}}(\text{EWBG}_{\text{SM}}) = 0.$$

Theorem 16.9 (First-order richer-scope requirement). *Any first-order electroweak phase transition for the measured Higgs sector requires $D_{\text{HV},+}$ and a richer-scope certificate $\mathcal{C}_{\text{HV},+}$.*

Theorem 16.10 (ZPE offset obstruction). *The absolute Higgs effective-potential offset is not a cosmological-constant prediction without a separate gravitational/source-ledger certificate.*

17 Claim and nonclaim register

ID	Statement	Status
HV-T1	Higgs-vacuum endpoint use instantiates the AASC kernel.	AASC theorem
HV-T2	Endpoint objects require typed quotient construction: $[V_{\text{eff}}]_{T=0}^{\text{stat}}, [\Gamma_{\text{decay}}]_{T=0}^{\text{life}}, [\mathcal{F}_T]_{T>0}^{\text{therm}}$.	Core theorem
HV-T3	Gauge-fixed field coordinates and instability scales are not primitive endpoints.	Nielsen/gauge theorem
HV-T4	RG crossings and running-coupling values are diagnostics, not endpoint selectors.	RG transport theorem
HV-T5	Top-mass direct-reconstruction anchors require mass-scheme/matching certificates.	Anchor theorem
HV-T6	The effective potential is admissible diagnostic content, not raw endpoint standing.	Positive admission theorem
HV-T7	The candidate primitive lower Higgs vacuum C_{lower} has no same-scope survivor.	Main false-vacuum endpoint
HV-T7a	The lower effective-potential branch has diagnostic availability, $\text{DiagnosticAvailability} = 1$, but not primitive endpoint standing, $\text{PrimitiveStanding} = 0$.	Non-promotion theorem
HV-T8	Stable/metastable selection requires an anchor-closed singleton certificate.	Burden theorem
HV-T9	Source-state anchors imply precision-sensitive zero-temperature audit, not structural singleton.	Certificate-failure theorem
HV-T10	The measured minimal SM electroweak transition is crossover.	Positive physics endpoint
HV-T11	First-order EWPT requires richer scope.	Same-scope deletion theorem
HV-T12	Same-domain route availability for C_{meta} factors through $\mathcal{R}_{\text{life}}$ and does not imply standing closure.	Route-availability exhaustion theorem
HV-T13	Same-scope measured minimal-SM EWBG route has no certified survivor.	Route-deletion theorem
HV-T14	BSM EWBG routes require richer-scope certificate $\mathcal{C}_{\text{EWBG},+}$.	Update theorem
HV-T15	Absolute Higgs potential offset is not a cosmological prediction.	ZPE quotient theorem
HV-N1	The paper does not compute a new instability scale.	Explicit nonclaim
HV-N2	The paper does not prove absolute stability.	Explicit nonclaim
HV-N3	The paper does not prove metastability as a structural singleton.	Explicit nonclaim
HV-N4	The paper does not rule out BSM first-order transitions.	Explicit nonclaim
HV-N5	The paper does not predict $\text{top/Higgs}/\alpha_s$ values.	Explicit nonclaim
HV-N6	The paper does not rule out richer-scope electroweak baryogenesis.	Explicit nonclaim

18 Same-target closure, weakening resistance, and cost of denial

18.1 Cost of kernel denial

Lemma 18.1 (Cost of kernel denial). *A denial of the AASC kernel while retaining the same Higgs-vacuum endpoint target either concedes equivalent governance work under different vocabulary or abandons a target-adequacy condition. Abandoning the condition incurs the corresponding loss in Table 6.*

Table 6: Cost of kernel denial in $D_{\text{HV},0}$.

Denied condition	adequacy	Kernel role lost	Cost to the Higgs-vacuum target
Target determinacy		Reference	The object drifts among raw potential, field coordinate, running coupling, lifetime class, thermal transition, anchor input, and richer-scope extension.
Status evaluability		Standing	Availability, diagnostic use, endpoint standing, declared input, empirical witness, and target update collapse into one undifferentiated status.
Act-time finality		Irreversibility	A failed same-scope branch can be repaired after audit by importing BSM carriers, new thresholds, gravity, or a narrower top-mass conversion while still being counted as the same endpoint act.
Same-regime fidelity		Admissibility	Gauge redescription, RG transport, scheme change, matching update, scalar extension, and UV completion become interchangeable rather than typed operations.

Proof. If a critic preserves the fixed target, distinguishable statuses, no-repair condition, and same-scope transport boundaries, then the critic has preserved the governance work named by the kernel. If a critic denies one of those conditions, the table records the target-level loss. Any later device that restores the lost good performs governance-equivalent work or changes target. $\square$

18.2 Denial obligation register

A denial of the Higgs endpoint theorem has standing only by discharging at least one load-bearing obligation:

1. show that endpoint use does not require target determinacy, status evaluability, act-time finality, or same-regime fidelity;
2. break the construction of $D_{\text{HV},0}^{T=0}[\text{S}_{2026}]$, $D_{\text{HV},0}^{T>0}$, or $D_{\text{EWBG},0}$;
3. show that C_{lower} , ν_{stat} , ν_{life} , ν_{therm} , or $D_{\text{HV},+}$ is ill-formed;
4. break effective-potential diagnostic admission;

5. produce a genuine fifth lower-branch endpoint function $F_5(B)$;
 6. break the candidate-route classifier, route-status completeness theorem, or lifetime-route normal-form exhaustion theorem;
 7. break the imported physical-Higgs crossover theorem;
 8. break the imported measured minimal-SM EWBG failure theorem;
 9. show a same-scope first-order branch survives inside $D_{\text{HV},0}^{T>0}$;
 10. show EWBG_{SM} survives inside $D_{\text{EWBG},0}$;
 11. supply a certified richer-scope package and accept target update; or
 12. derive the kernel roles from below without importing governance-equivalent work.
- Objections outside these obligations have no same-target force.

18.3 Core Higgs-vacuum adequacy structure

Definition 18.2 (Core Higgs-vacuum adequacy structure). The structure $\mathcal{C}_{\text{HV}}^0$ consists of $D_{\text{HV},0}$, the typed split $D_{\text{HV},0}^{T=0}/D_{\text{HV},0}^{T>0}$, the kernel roles, the admissibility-relevant invariant bundle I_D including $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ for lifetime claims, gauge-vacuum compatibility, RG/scheme/matching transport, top-mass anchor discipline, the verdict maps ν_{stat} , ν_{life} , ν_{therm} , the typed quotients $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, $[\mathcal{F}_T]_{T>0}^{\text{therm}}$, the singleton certificate burden $\mathcal{S}_{\text{HV}}^{T=0}(C)$, the richer-scope update rule, and the endpoint-selection criterion.

Definition 18.3 (Endpoint-permissive weakening). An endpoint-permissive weakening is a purported weaker regime that preserves $\mathcal{C}_{\text{HV}}^0$ while allowing one of the following without the corresponding burden: primitive standing for a raw lower branch, same-scope first-order electroweak transition in the measured minimal SM, or zero-temperature stable/metastable singleton selection without $\mathcal{S}_{\text{HV}}^{T=0}(C)$.

Theorem 18.4 (No strict same-target weakening permits the forbidden promotions). *No strict weakening preserves $\mathcal{C}_{\text{HV}}^0$ while permitting primitive lower-branch standing, same-scope first-order electroweak transition in $D_{\text{HV},0}^{T>0}$, or zero-temperature singleton selection without a closed singleton certificate.*

Proof. Exhaust the weakening routes. If reference is weakened, the claim is no longer about the same Higgs-vacuum target. If standing is weakened, diagnostic availability becomes endpoint selection and the theorem no longer evaluates claim-bearing endpoint status. If admissibility is weakened, gauge changes, RG/scheme moves, top-mass conversion, and richer-scope updates collapse into one operation. If irreversibility is weakened, a failed branch can be repaired by later hidden carrier import. If the weakening adds a middle status, that status either affects endpoint selection, in which case it is hidden selector work, or it is inert bookkeeping. If all members of $\mathcal{C}_{\text{HV}}^0$ are preserved, the lower-branch, singleton-burden, crossover, and richer-scope theorems still apply. Therefore the weakening either changes target, imports governance-equivalent work, becomes inert, or preserves the endpoint theorem. $\square$

18.4 No cost-free dependence objection

AASC dependence fixes the theorem class and is not a defect. The dependence would be defective only if an imported kernel role were false, a source theorem were misapplied, a target branch were

smuggled into a definition, or a same-target counterexample preserved the target while avoiding the burden. A refusal to use the word “standing” while still distinguishing representative, diagnostic, endpoint, input, witness, and target update concedes the same work under different terminology.

19 Same-target closure and objection discipline

19.1 Standing-objection register

Objection	Standing	Response
Effective potentials show a lower branch.	Partial	Granted: DiagnosticAvailability = 1. Endpoint standing requires ν_{stat} , ν_{life} , ν_{therm} , or $D_{\text{HV},+}$.
Standard analyses discuss metastability.	Granted	Metastability is a certificate-dependent endpoint class routed through ν_{life} , not primitive lower-branch standing. Standing closure requires the singleton/lifetime certificate to close.
Standard minimal-SM calculations yield long-lived metastability without new physics.	Granted as a ν_{life} -route; non-refuting as a primitive-endpoint objection	Such calculations may certify C_{meta} when I_D , $\text{Cont}_{\text{SM}}(\Lambda\mathcal{P})$, and the singleton/lifetime certificate close. They do not certify primitive endpoint standing for C_{lower} .
A theoretical metastability route exists in the minimal SM.	Granted as route availability; non-refuting as standing closure	It is classified as $\mathcal{R}_{\text{life}}$ -route availability, not endpoint standing.
Standard RG plus bounce pipeline should count as the lower-vacuum endpoint.	No as endpoint-standing shortcut	The pipeline is route-normal only if I_D , Cont_{SM} , $\mathcal{L}$, U_Δ , and the source-state components are declared and supplied. It becomes standing-closed only if the singleton/lifetime certificate closes.
Route availability should be enough for endpoint use.	None	This is standing-smuggling; route availability is not a generator, carrier, or repair of standing.
AASC terminology is refused.	None	Refusal of terminology does not break a theorem link; a same-target objection must attack a load-bearing definition, quotient, verdict map, or inference.
A new numerical instability-scale calculation is required.	None unless certificate-relevant	A recalculation matters only if it supplies, breaks, or updates a declared verdict certificate.

Gauge-fixed instability scales are useful.	Partial	Useful as diagnostics; not primitive endpoint objects without gauge-controlled verdict descent.
Top mass is the key uncertainty.	Granted	The top-mass anchor non-identity theorem and singleton certificate make it a load-bearing component.
BSM can make the transition first-order.	Compatible	That is $D_{\text{HV},+}$, a richer-scope update, not a same-scope counterexample to $D_{\text{HV},0}^{T>0}$.
EWBG may work in extended models.	Compatible	That is $D_{\text{EWBG},+}$, not survival of EWBG_{SM} inside $D_{\text{EWBG},0}$.
Future precision may select stability or metastability.	Compatible	That is a source-state update $\mathbf{S}_{2026} \rightarrow \mathbf{S}_+$, not a refutation of the present source-state theorem.
The lower branch should count as physical by convention.	None	Convention is not endpoint standing unless it supplies a verdict map, singleton certificate, or source-continuity package.
A fifth endpoint function exists.	Serious if specified	The objector must supply $F_5(B)$ preserving the same target and not reducing to static, lifetime, thermal, or richer-scope update.

19.2 Faithful same-regime counterexample conditions

A faithful same-target counterexample must preserve:

1. $D_{\text{HV},0}$;
 2. the measured Higgs/top/electroweak/ α_s anchor ledger;
 3. the typed target split $D_{\text{HV},0}^{T=0}$ versus $D_{\text{HV},0}^{T>0}$;
 4. the admissibility-relevant invariant bundle I_D , including $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ for lifetime claims;
 5. the lifetime-route normal form $\mathcal{R}_{\text{life}}$ for any claimed same-domain route to C_{meta} ;
 6. gauge-vacuum compatibility;
 7. RG/scheme/matching transport discipline;
 8. top-mass anchor conversion discipline;
 9. the typed quotients $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, and $[\mathcal{F}_T]_{T>0}^{\text{therm}}$;
 10. absence of richer-scope carrier import;
 11. the endpoint selection criterion;
- while producing at least one of:
1. a same-scope first-order electroweak phase transition in the measured minimal SM;
 2. structural stable/metastable singleton selection without $\mathcal{S}_{\text{HV}}^{T=0}(C)$;
 3. primitive endpoint standing for a raw lower effective-potential branch independent of quotient audit;
 4. a same-target route bypassing top-mass and matching conversion while preserving RG endpoint selection.

A construction that adds a scalar, threshold, symmetry, UV rule, gravitational source, empirical witness, or altered high-scale continuation clause is a target update, not a same-target counterexample.

19.3 Right-mode objection targets

A same-target objection has standing only if it attacks at least one load-bearing link below:

1. target adequacy to kernel roles;
2. context non-smuggling in the definition of $D_{\text{HV},0}$;
3. construction of verdict maps and quotient kernels;
4. gauge-vacuum compatibility and Nielsen/Fukuda-Kugo use;
5. RG/scheme/matching transport discipline;
6. I_D and $\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$ for lifetime-route claims;
7. route-availability exhaustion, the lifetime-route normal-form theorem $\mathcal{R}_{\text{life}}$, or a genuine fifth same-domain route-availability class not reducible to diagnostic-only, route-incomplete, route-normal, target-update, or standing-smuggling;
8. top-mass anchor non-identity;
9. effective-potential diagnostic admission versus endpoint promotion;
10. lower-branch non-promotion;
11. singleton-certificate burden;
12. rival-basin non-deletion;
13. source-state certificate failure;
14. imported finite-temperature crossover theorem;
15. same-scope first-order deletion;
16. richer-scope update normal form;
17. ZPE absolute-offset obstruction;
18. final endpoint-selection inference; or
19. a genuine fifth lower-branch endpoint function satisfying Criterion 7.25.

Objections outside these classes change the target or leave the theorem chain intact unless they identify a theorem-level failure in one of the cited links.

20 Conclusion

The lower Higgs vacuum is not a raw lower-potential fact. It is a quotient endpoint-status problem. A field-coordinate lower branch, a gauge-fixed instability scale, a running quartic crossing, or an absolute potential offset cannot be promoted directly into the fate of the electroweak vacuum. The endpoint object must be a class in $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, or $[\mathcal{F}_T]_{T>0}^{\text{therm}}$, or a declared richer-scope update. The lower branch remains an admitted diagnostic; what is deleted is primitive endpoint standing. Same-domain route availability for metastability is likewise exhausted in form: a route to C_{meta} must factor through $\mathcal{R}_{\text{life}}$. Candidate metastability protocols are first classified by ρ_{route} ; only

RouteNormal protocols become route-available through NF_{life} ; and only certificate closure turns route availability into endpoint standing. In plain terms, a calculation may be a candidate, may become a route, and may still fall short of a standing endpoint until the required certificate closes. Route availability alone does not close standing, and future closure of that route selects C_{meta} , not primitive standing for C_{lower} .

The strongest same-scope endpoint is finite-temperature. In the measured minimal Standard Model, the electroweak transition is crossover:

$$\text{Sel}_{D_{\text{HV},0}^{T>0}}(C_{\text{cross}}) = 1.$$

The first-order branch is deleted inside the same target:

$$\text{Surv}_{D_{\text{HV},0}^{T>0}}(C_{\text{1st}}) = 0,$$

and re-enters only through $D_{\text{HV},+}$ with a richer-scope certificate.

The zero-temperature endpoint is different. Absolute stability and metastability are legitimate endpoint classes. But selecting either as a fixed-scope structural singleton requires $\mathcal{S}_{\text{HV}}^{T=0}(C)$, a certificate closing gauge, RG/scheme, top-mass conversion, matching, perturbative-order, uncertainty, high-scale continuation, and lifetime components. Source-state anchors do not by themselves provide that certificate. Thus the zero-temperature audit verdict is precision-sensitive endpoint occupation, not raw lower-branch promotion and not structural singleton selection. Standard minimal-SM metastability calculations remain compatible with this result: when they declare the high-scale invariant bundle and close the lifetime certificate, they license C_{meta} through ν_{life} rather than primitive standing for C_{lower} .

This conclusion is the strengthened endpoint: it is protected by explicit quotient construction, an explicit singleton burden, a top-mass anchor theorem, a positive effective-potential admission theorem, and a same-scope crossover theorem.

The endpoint is AASC-internal in the precise necessity sense: any claim-bearing endpoint use of the fixed measured Higgs target already requires reference, standing, admissibility, and irreversibility. A contrary same-target construction must derive a kernel role from below, break a load-bearing definition or theorem link, produce a genuine fifth lower-branch endpoint function, identify a source-audit failure, or satisfy the faithful same-target counterexample conditions. Without one of those results, refusing AASC vocabulary either concedes the same governance work under another name or abandons the fixed Higgs-vacuum endpoint target.

A Forbidden-promotion register

Input		Forbidden promotion	Required certificate or endpoint
Lower branch of V_{eff}		Primitive lower-vacuum endpoint	Diagnostic availability is admitted; endpoint standing requires ν_{stat} , ν_{life} , ν_{therm} , or $D_{\text{HV},+}$.
$\lambda(\mu) < 0$		Selected metastability endpoint	Singleton and lifetime certificate $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}})$.
Gauge-fixed scale	instability	Physical endpoint scale	Gauge-controlled verdict map.

Effective-potential offset	Cosmological-constant prediction	Source/gravity ledger certificate; otherwise ZPE obstruction.
First-order BSM thermal history	Same-scope SM first-order survivor	$D_{\text{HV},+}$ certificate.
SM crossover	All baryogenesis impossible	Only same-scope measured minimal-SM EWBG route deletion follows.
CKM CP phase	Adequate EWBG source	Imported source adequacy and transport certificate.
Future precision input	Contradiction of current theorem	Source-state update $S_{2026} \rightarrow S_+$.
Unknown stabilizer or UV repair	Same-scope survival	Specified carrier, law, boundary condition, descent map, and witness package.

B Anti-circularity and forbidden-retyping audit

Potential circle or forbidden retyping	Audit response
$D_{\text{HV},0}$ smuggles in crossover or non-singleton status.	The target fixes carrier inventory, anchors, and comparison arenas only. Crossover and non-singleton audit follow from later theorem blocks.
The quotient is defined by the desired endpoint.	Verdict maps ν_{stat} , ν_{life} , and ν_{therm} are stated first; quotient relations are their same-scope kernels.
A lower branch is deleted by definition.	It is admitted as diagnostic availability and deleted only as primitive endpoint standing after verdict-map and no-fifth-function exhaustion.
C_{lower} is defined as unsupported.	False. It is the neutral primitive lower-branch claim-form under test; unsupported status is the theorem output.
The no-fifth-function theorem is only taxonomy.	It is an endpoint-function exhaustion theorem. A standing objection must produce $F_5(B)$ preserving the same target and not reducing to static, lifetime, thermal, or richer-scope update.
Metastability is weakened out of the paper.	Metastability remains a legitimate endpoint class; it requires $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, and $\mathcal{S}_{\text{HV}}^{T=0}(C_{\text{meta}})$ for selection.
Crossover is assumed by AASC.	Crossover is imported as the standard physical-Higgs SM finite-temperature result; AASC classifies its endpoint status inside $D_{\text{HV},0}^{T>0}$.
First-order transitions are ruled out globally.	They are ruled out only as same-scope minimal-SM endpoints and re-enter through $D_{\text{HV},+}$ with carrier certificates.
Top mass is ignored.	Top-mass interpretation is a central anchor-transport burden and a component of $\mathcal{S}_{\text{HV}}^{T=0}(C)$.

ZPE obstruction erases physical vacuum-energy differences.	Only absolute offset promotion is blocked; relational phase/free-energy/tunneling/lifetime quantities survive.
AASC dependence makes the result optional.	Same-target endpoint use already needs target identity, status evaluability, no-repair, and same-regime fidelity.
	Denying them has the costs recorded in Table 6.

C AASC endpoint-architecture source lineage

The proof architecture of this manuscript follows the established endpoint posture of two AASC source manuscripts. The Strong-CP endpoint paper supplies the quotient-survivor, selector-burden, no-one-step-retyping, cost-of-kernel-denial, no-strict-weakening, faithful-counterexample, right-mode-objection, and anti-circularity pattern. The Neutrino-Mass endpoint paper supplies the target-fixation-before-branch-evaluation, availability/occupancy/survival/selection, low-energy quotient, carrier/protector burden, no-one-step-retyping, cost-of-denial, weakening-resistance, and faithful-counterexample pattern. This manuscript imports that architecture only at the level of proof posture and status calculus. It does not import the strong-CP or neutrino endpoint conclusions.

Recent manuscript posture	Transferred proof role	Higgs-vacuum specialization
Mathematical-status lock	Declared proof class; dependence is evaluated by theorem links, not by vocabulary preference	Section 2 states AASC as fixed-target status calculus over standard Higgs/QFT arena.
Dependency-inversion exclusion	Candidate branch cannot define the standing rule by which it wins	Effective-potential lower branch, RG crossing, and first-order thermal claim cannot self-select.
No one-step retyping	Prevents shortcut from witness to endpoint	Lower branch and $\lambda(\mu) < 0$ must pass through $[V_{\text{eff}}]_{T=0}^{\text{stat}}$, $[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$, and $\mathcal{S}_{\text{HV}}^{T=0}(C)$.
Cost of kernel denial	Denying kernel roles incurs target-level losses	Table 6 records the corresponding losses for Higgs-vacuum claims.
No strict same-target weakening	Blocks weakening that preserves target while admitting forbidden branch	$\mathcal{C}_{\text{HV}}^0$ theorem blocks lower-branch promotion, same-scope first-order transition, and singleton selection without certificate.
Faithful counterexample conditions	Specifies what a real same-regime counterexample must preserve	Section 19.2 fixes preservation conditions and allowed contrary outputs.
Right-mode objection targets	Separates standing objections from target-changing objections	The objection register names the load-bearing links that must be attacked.
Anti-circularity audit	Excludes definition-by-conclusion and representative retyping	Appendix audit records context non-smuggling, quotient-kernel construction, and false-vacuum burden.

D Source-lock and dependency map

Source family	Imported role	Use in this manuscript
Gauge-Vacuum Compatibility	Vacuum standing, gauge-fixing non-promotion, vev witness discipline	Blocks raw gauge-fixed extrema, vevs, and field-coordinate minima from primitive endpoint use.
Higgs-Like Bridge and Scale-Anchor Rigidity	Higgs bridge carrier, scale-anchor discipline, no vev/potential/RG-chart selector	Prevents Higgs notation, bridge normalization, scalar-potential coefficients, or running bridge labels from selecting endpoints by themselves.
Renormalization as Standing-Preserving Scale Transport	RG as transport, scheme skin, effective-envelope/matching discipline	Classifies $\lambda(\mu)$ crossings and RG-chart features as diagnostics requiring endpoint transport.
The Higgs Bridge and the Metric Envelope	Higgs bridge/fixation carrier versus vev, pole mass, potential coordinate, RG chart value	Supports endpoint-status caution and no substitution of representative for bridge-standing content.
Standalone Standard Model Structure	Minimal SM structural target and parameter-program boundary	Fixes carrier inventory while keeping numerical Higgs/top/coupling values as measured anchors.
Strong-CP Boundary Endpoint	Quotient survivor discipline, availability/occupancy distinction, vacuum-energy profile non-selector, richer-scope normal form	Provides endpoint architecture for boundary load, source continuity, and target-update classification.
Neutrino-Mass Low-Energy Endpoint	Target fixation before branch evaluation, branch-status calculus, low-energy quotient rule	Provides status calculus and no-one-step-retyping architecture.
ZPE trilogy	Absolute zero-point offset obstruction, relational survivor classification, renormalization-standing equivalence	Blocks absolute effective-potential offset as cosmological prediction while preserving relational phase/tunneling/free-energy differences.
External Higgs/top/vacuum-stability/EWBG literature	Measured anchors, high-order vacuum-stability dependence, gauge-dependence, crossover, CKM insufficiency, and BSM facts	Supplies standard physics comparison arena.

E Failure-route table

Route	Attempted promotion	Audit result
Gauge-fixed minimum	Field-coordinate extremum becomes endpoint	Deleted unless quotient-stable or mapped to physical predicate.

Instability scale	Field value where V_{eff} crosses below EW value becomes physical scale	Gauge/scheme diagnostic; not raw endpoint.
$\lambda(\mu) < 0$	Running quartic sign becomes metastability proof	Diagnostic requiring matching to lifetime/stability classification.
Lower plotted branch	Lower potential region becomes selected false vacuum	Non-promoted until gauge/RG/matching/lifetime audit closes.
Direct top mass input	Monte-Carlo mass parameter becomes RG mass without conversion	Anchor-transport failure unless mass-scheme certificate supplied.
Absolute potential offset	Higgs potential value becomes cosmological constant	ZPE quotient obstruction; no standalone prediction.
First-order EWPT in SM	Baryogenesis-friendly thermal transition inside minimal measured SM	Same-scope deletion; requires richer scope.
SM EWBG	Minimal SM supplies both first-order dynamics and adequate CP source	Route deletion; first-order and CP/source limbs both fail.
Zero-temperature metastability	Lower branch or metastability rescues EWBG	Route laundering failure; thermal and CP/source ledgers remain missing.
Singlet scalar repair	New scalar stabilizes or makes transition first-order while counted as same target	Target update $D_{\text{HV},+}$.
Unknown UV stabilizer	Unspecified high-field stabilizer	No source-continuity work unless specified; if specified, richer scope.

F External physics citation map

External source	Role in manuscript
ATLAS Higgs mass [1]	Fixes Higgs-mass anchor $m_H = 125.11 \pm 0.11$ GeV.
ATLAS+CMS top mass [2]	Fixes source-state direct-reconstruction top-mass anchor $m_t = 172.52 \pm 0.33$ GeV.
Hoang top-mass review [3]	Supports top-mass interpretation and Monte-Carlo mass versus field-theory mass-scheme burden.
Degrassi et al. [4]	Canonical NNLO Standard Model Higgs-vacuum stability analysis.
Buttazzo et al. [5]	Canonical near-criticality and high-scale stability/metastability map.
Hiller et al. [6]	Supports source-state top-mass/ α_s load-bearing uncertainty and precision requirement.
Fukuda-Kugo/Nielsen identity literature [8, 7]	Supports gauge-dependence mechanics of the effective potential.
Di Luzio-Mihaila [9]	Supports gauge-dependent instability field value versus gauge-independent stability bound distinction.

Andreassen-Frost-Schwartz [10]	Supports consistent gauge/order use of the Standard Model effective potential.
Coleman and Callan-Coleman [11, 12]	Support bounce/action decay-rate structure for false-vacuum lifetime status.
D’Onofrio-Rummukainen [13]	Supports physical-Higgs SM crossover and $T_c = 159.5 \pm 1.5$ GeV.
Laine-Meyer [14]	Supports thermodynamics across electroweak crossover and no bulk transition.
Schicho-Tenkanen-White [15]	Supports modern gauge-invariant/resummed electroweak phase-transition computation concerns.
Camargo-Molina-Enberg-Lofgren [16]	Example of richer-scope first-order EWPT/SMEFT catalog.
Cohen-Kaplan-Nelson [17]	Canonical EWBG review fixing route conditions and early repair logic.
Morrissey-Ramsey-Musolf [18]	Modern EWBG review supporting strong-transition and new CP/source repair burden.
Huet-Sather [19]	Supports CKM-only CP insufficiency for electroweak baryogenesis.
Huet [20]	Supports measured minimal-SM EWBG failure and richer-scope repair by extended symmetry-breaking sector plus new CP source.

G Notation and glossary

Symbol	Meaning
$D_{\text{HV},0}$	Fixed measured minimal Standard Model Higgs-vacuum target.
$D_{\text{HV},0}^{T=0}$	Zero-temperature stability/metastability/lifetime target.
$D_{\text{HV},0}^{T>0}$	Finite-temperature electroweak-transition target.
$D_{\text{HV},+}$	Richer-scope target after new carrier, threshold, UV, symmetry, or gravitational load.
$D_{\text{EWBG},0}$	Same-scope measured minimal-Standard-Model EWBG target for producing and preserving the observed baryon asymmetry.
$D_{\text{EWBG},+}$	Richer-scope electroweak-baryogenesis update target.
$[V_{\text{eff}}]_{T=0}^{\text{stat}}$	Zero-temperature static effective-potential diagnostic quotient.
$[\Gamma_{\text{decay}}]_{T=0}^{\text{life}}$	Zero-temperature decay/lifetime quotient.
$[\mathcal{F}_T]_{T>0}^{\text{therm}}$	Finite-temperature thermodynamic transition quotient.
$S_{\text{HV}}^{T=0}(C)$	Anchor-closed singleton certificate for zero-temperature endpoint C .
C_{abs}	Absolute-stability endpoint class.
C_{meta}	Metastability endpoint class.
C_{unst}	Instability endpoint class.
C_{lower}	Candidate claim that a lower branch itself has primitive same-scope endpoint standing; deleted by route exhaustion.
C_{cross}	Smooth electroweak crossover class.
$C_{1\text{st}}$	First-order electroweak phase-transition class.

A_{HV}	Measured Higgs/top/electroweak/strong-coupling anchor ledger.
I_D	Admissibility-relevant invariant bundle fixing same target, same standing-bearing status, and same admissibility boundary.
$\text{Cont}_{\text{SM}}(\Lambda_{\mathcal{P}})$	High-scale minimal-SM continuation clause included in I_D for lifetime verdicts.
$\mathcal{R}_{\text{life}}$	Same-domain lifetime-route normal-form tuple for C_{meta} ; $\text{NF}_{\text{life}}(\mathcal{P})$ is the partial normalizer defined only for route-normal protocols.
RouteAvail	Route-availability predicate; it does not imply standing closure.
StandingClosed	Endpoint standing closure after the declared verdict and certificate close.
U_{Δ}	Uncertainty and rival-basin separation envelope.
Availability	Formal writability or diagnostic availability inside the declared arena.
Standing	Endpoint-usable status after quotient, transport, and admissibility filters.
Selection	Unique certified survivor quotient under the endpoint criterion.
Richer scope	Target update by added carrier, law, threshold, witness, or boundary input.
No-promotion	Failure of representative, diagnostic, or boundary data to become endpoint standing without certificate.
$\mathcal{C}_{\text{EWBG},+}$	Richer-scope electroweak-baryogenesis certificate.

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